

A Virtual Yeast Cell for Metabolic Engineering: Genome-Scale Flux Balance Analysis, Graph Deep Learning, and Isozyme Over-Rescue: an Expression-Blindness Failure of Genome-Scale Models, Discovered in Yeast and Replicated Across Three Species

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Abstract

We build a virtual *Saccharomyces cerevisiae* cell on the consensus genome-scale metabolic model yeast-GEM (4,105 reactions, 2,748 metabolites, 1,143 genes) and subject it to a locked, gene-held-out evaluation protocol against the published gene-essentiality benchmark. Three contributions result. First, we reproduce the published benchmark exactly (accuracy 0.9015, MCC 0.532) and show that on thresholded classification no learned model beats the FBA knockout rule (GCN $\Delta\text{MCC} = -0.017$, $\text{CI}_{95} [-0.052, 0.022]$), an honest negative we preserve and analyse. Second, on threshold-free ranking the same graph convolutional network breaks the benchmark decisively: AUC-ROC 0.831 vs 0.697 ($\Delta\text{AUC} = +0.135$, $\text{CI}_{95} [0.114, 0.155]$, paired bootstrap $p < 2 \times 10^{-4}$). Third, we discover and quantify a named failure mode of the state-of-the-art model, *isozyme over-rescue*: genes whose essentiality the model misses carry alternative-isozyme clauses in its gene-protein-reaction rules at 20.2% versus 1.5% for genes it catches (odds ratio 16.2, Fisher $p = 3.8 \times 10^{-4}$); we name the 19 affected genes and ship a reusable auditor. The phenomenon generalises: the same audit reproduces it in *E. coli* (iML1515 vs the Keio collection; OR 11.5, $p = 1.2 \times 10^{-5}$) and *B. subtilis* (iYO844 vs the Koo 2017 libraries; OR 22.7, $p = 2.5 \times 10^{-7}$). Protein-abundance data (PaxDb) show the model-trusted backups are expressed far below the essentials they supposedly rescue in yeast ($p = 0.038$) and *E. coli* ($p = 0.007$), and curated genetics (BioGRID) shows overexpressed backups *do* rescue—the failure is expression-blindness, not wrong chemistry. In *B. subtilis* the abundance mechanism does *not* replicate ($p = 0.29$), an honest negative we analyse. Finally we run a full 1,143-gene in-silico knockout screen for succinate overproduction, validate against the published quadruple-deletion strain of Raab et al. (2010), and nominate novel, falsifiable strain-design candidates.

Contents

1	Introduction	4
1.1	Why essentiality is the right benchmark	4
1.2	Why strain design is the right application	4

2	Related work	4
2.1	Genome-scale models of yeast	4
2.2	Machine learning on metabolic networks	5
2.3	Strain-design algorithms	5
3	Data and reproducibility	5
3.1	The metabolic model	5
3.2	Essentiality gold standard	5
3.3	Protein sequences	5
3.4	Media	5
3.5	External validation datasets	6
4	Methods	6
4.1	Flux balance analysis	6
4.2	Gene essentiality screen	6
4.3	Label-free mechanistic features	6
4.4	Graph convolutional network	6
4.5	Sequence CNN	7
4.6	GPR parsing and isozyme redundancy	7
4.7	Hyperparameters and training	7
4.8	Strain-design metrics	7
4.9	Locked evaluation protocol	7
4.10	Medium dependence as a first-class variable	8
5	Mathematical results	8
6	Results	10
6.1	The published benchmark reproduces exactly	10
6.2	Locked cross-validation: the MCC gate fails honestly	10
6.3	The GCN breaks the benchmark on ranking	10
6.4	Threshold instability explains the MCC–AUC split	10
6.5	Discovery: isozyme over-rescue	11
6.6	The nineteen over-rescued genes	14
6.7	Strain design for succinate overproduction	15
6.8	Essentiality triangulation and a curation conflict	19
6.9	External experimental validation	19
7	Negative results and failed approaches	19
7.1	Synthesis: a four-link mechanism chain, six species	20
7.2	Cross-study replication: the DEG panel	20
7.3	Per-study abundance replication; a tempered yeast claim	21
7.4	Co-expression and evidence channels: annotation transfer in yeast	22
7.5	Enrichment and structural coverage: scattered generic enzymes	22

8	Discussion	22
8.1	Limitations	23
8.2	Future work	23
9	Per-model fold analysis	23
10	Conclusion	23
A	Reproducibility	25
B	Feature definitions	25
C	Strain-design tables	25
D	Per-gene essentiality scores	25
E	Full tool output: all benchmark genes	29
F	External tools and dataset manifest	57

1 Introduction

Genome-scale metabolic models (GEMs) are the working mathematical description of cellular metabolism. Flux balance analysis (FBA) turns such a model into a linear program whose optimal objective value predicts growth, and single-gene-deletion FBA is the standard in-silico essentiality screen. The consensus yeast model yeast-GEM reports 90.15% essentiality accuracy on synthetic complete medium, which corresponds to a Matthews correlation coefficient (MCC) of 0.532: 65 of 159 observed essential genes are recovered, 94 are missed, and 15 viable genes are falsely predicted essential. This project asks two questions. (i) Can machine learning on model structure and protein sequence beat that benchmark under a protocol that never lets a test gene influence training? (ii) When the model is wrong, is it wrong systematically—and can that system be named, quantified and audited? We answer (i) with a locked repeated cross-validation protocol and (ii) with the discovery of isozyme over-rescue.

1.1 Why essentiality is the right benchmark

Gene essentiality is the rare benchmark in systems biology where the ground truth is nearly unambiguous: a deletion either grows or it does not, under a defined medium, and the community has measured it at genome scale with the same strain background. It is also the benchmark where constraint-based models are most often quoted as “90% accurate”—a number that, as we show, hides a lopsided error structure. Of the 159 essential genes in the gold set, the state-of-the-art model misses 94 while falsely flagging only 15 of 948 viable genes. Accuracy rewards the easy majority class; MCC and AUC do not.

1.2 Why strain design is the right application

Essentiality screening asks the model to say where the cell breaks. Strain design asks the opposite: where can the cell be broken *usefully*? Succinate is a top-value bio-based chemical, and *S. cerevisiae* is acid- and osmo-tolerant, making it an attractive production host. The published record gives us something rare: a quadruple-knockout design (Raab et al., 2010) derived from classical TCA reasoning and validated experimentally, plus a genome-scale-model-driven study (Otero et al., 2013). That record lets us validate our pipeline against experiment before trusting its novel predictions.

2 Related work

2.1 Genome-scale models of yeast

The yeast metabolic reconstruction lineage runs from iFF708 and iND750 through the Yeast consensus series to yeast-GEM (Yeast8), which unified the community model with standardised annotations and an open versioning ecosystem. Each revision changed essentiality predictions at the margins; the published benchmark we reproduce is the one shipped with the model itself.

2.2 Machine learning on metabolic networks

Deep learning has been applied to flux prediction, to gene essentiality across bacteria and cancer cell lines, and to promoter or sequence function. Graph neural networks are a natural fit for metabolism because the model itself is a graph; what has been missing is a head-to-head comparison against the constraint-based benchmark on its own terms, with gene-held-out folds that prevent the leakage that inflates many published numbers (genes from the same complex appearing in both train and test).

2.3 Strain-design algorithms

OptKnock framed design as bilevel optimisation: the cell maximises growth, the engineer chooses deletions to make production obligatory at that optimum. Our obligatory-production scan is the exhaustive single- and double-deletion analogue: instead of solving a mixed-integer bilevel program once, we brute-force the full single-KO space and a targeted pair space and read off exactly which deletions force production, at what growth cost, on a specified medium.

3 Data and reproducibility

3.1 The metabolic model

The SBML fixture `data/raw/yeast-GEM.xml` (yeast-GEM, SysBioChalmers) is stored in the repository; its SHA-256 is recorded with every result. Statistics: 4,105 reactions, 2,748 metabolites, 1,143 genes.

3.2 Essentiality gold standard

The Stanford deletion-collection classifications (`data/raw/essentialGenes.tsv`) label 1,107 model genes. The file uses a viable-positive convention; we verified by two independent checks that observed-essential corresponds to its {TN, FP} classes (159 genes): all 159 appear in the SGD inviable-ORF list, and knockout growth ratios separate the classes exactly.

3.3 Protein sequences

Sequences for all 1,143 model genes were recovered from a Swiss-Prot yeast table by alias-token mapping (coverage 1,143/1,143).

3.4 Media

Kennedy synthetic complete medium (Y7) is imposed by exchange-reaction bounds identical to the published benchmark, so all numbers are directly comparable.

3.5 External validation datasets

Three published experimental datasets, stored verbatim under `data/raw/`, validate the model outside the essentiality task: the aerobic glucose-limited chemostat compendium of Tobias 2013 (32 steady states with measured uptake and growth rates), the anaerobic chemostat study of Van Hoek 1998 (4 steady states with growth and CO₂/glucose ratios), and the Biolog substrate panel (383 substrates with experimental growth/no-growth calls; one accession-level dataset, 383 conditions).

4 Methods

4.1 Flux balance analysis

FBA solves

$$\begin{aligned} \max_v \quad & c^\top v \\ \text{s.t.} \quad & Sv = 0, \quad l \leq v \leq u, \end{aligned} \tag{1}$$

where $S \in \mathbb{R}^{m \times n}$ is the stoichiometric matrix and c selects the biomass reaction. Parsimonious FBA (pFBA) adds a second stage:

$$\min_v \|v\|_1 \quad \text{s.t.} \quad Sv = 0, \quad l \leq v \leq u, \quad c^\top v = c^\top v^*, \tag{2}$$

selecting the minimum-total-flux vector among optimal growth states. Flux variability analysis (FVA) computes, per reaction i ,

$$v_i^{\min} = \min\{v_i : Sv = 0, \quad l \leq v \leq u, \quad c^\top v \geq \gamma c^\top v^*\} \tag{3}$$

and $v_i^{\max}$ symmetrically, with $\gamma = 0.9$ here.

4.2 Gene essentiality screen

For each gene g we delete it (zero bounds on all reactions whose GPR requires g) and re-solve (1); the predicted viability ratio is $\rho_g = c^\top v_{-g}^* / c^\top v^*$.

4.3 Label-free mechanistic features

Eleven features per gene, computed from model structure only: knockout growth ratio; reaction count; minimum isozyme-clause count over its GPRs; maximum complex size; sole-reaction count; maximum and total pFBA flux carried; active-reaction fraction; gene-graph degree; compartment count; subsystem count. No feature consults the essentiality labels.

4.4 Graph convolutional network

Genes form a graph through shared non-currency metabolites (currency metabolites such as ATP, NADH, water are excluded; metabolites joining more than 60 genes are dropped as hubs). With symmetric normalised adjacency $\hat{A} = \hat{D}^{-1/2}(A + I)\hat{D}^{-1/2}$, propagation is

$$H^{(l+1)} = \text{ReLU}(\hat{A}H^{(l)}W^{(l)}), \quad H^{(0)} = X, \tag{4}$$

with two layers, hidden width 64, dropout 0.3, and a residual skip connection from X . Training is transductive-masked: features of all nodes are visible, but the loss reads training-fold labels only.

4.5 Sequence CNN

Protein sequences (truncated to 800 residues) are embedded and convolved with kernels of width 3, 7, 15, followed by masked global max pooling and a linear head (embedding 16, 32 channels, dropout 0.3).

4.6 GPR parsing and isozyme redundancy

Each reaction’s gene-protein-reaction rule is a Boolean formula in which `and` joins subunits of a required complex and `or` joins alternative ways to catalyse the reaction. We parse each rule into clauses $R = C_1 \vee C_2 \vee \dots \vee C_k$ where each C_i is a set of genes. For gene g we define

$$\text{minIso}(g) = \min_{R \ni g} k_R, \quad \text{soleRxns}(g) = \#\{R \ni g : k_R = 1\}, \quad (5)$$

so $\text{minIso}(g) = 1$ marks a gene with at least one reaction for which the model knows *no* alternative catalyst. Section 6.5 shows this single number predicts *where the model is wrong*.

4.7 Hyperparameters and training

All hyperparameters were fixed before the locked folds were generated; none were tuned on test outcomes. GCN: 2 layers, hidden width 64, dropout 0.3, Adam (lr 10^{-2} , weight decay 5×10^{-4}), 300 epochs, class-weighted BCE with $\text{pos_weight} = n_-/n_+$. CNN: embedding 16, three kernel widths $\{3, 7, 15\}$, 32 channels each, dropout 0.3, Adam (lr 3×10^{-3} , weight decay 10^{-4}), 25 epochs, batch 64. Logistic regression: balanced class weights, ℓ_2 penalty, max 2000 iterations. Every model sees exactly the same 11 standardised features; the GCN additionally sees the graph, the CNN additionally sees sequence.

4.8 Strain-design metrics

For a knockout set K and medium M , let $g^*(K)$ be maximal growth. The *obligatory production* is

$$\text{obl}(K) = \min\{v_p : Sv = 0, l_K \leq v \leq u_K, c^\top v \geq 0.999 g^*(K)\}, \quad (6)$$

i.e. the least succinate any near-optimal-growing cell must excrete. $\text{obl}(K) > 0$ is the FVA certificate of growth coupling; it is the quantity OptKnock-type designs maximise implicitly. We compute it exactly with two LPs per knockout set.

4.9 Locked evaluation protocol

Repeated stratified 5-fold cross-validation, 3 repeats, folds generated once per repeat from a fixed seed and shared by every model. Feature standardisation is fit on training folds only; classification thresholds are tuned on training-fold scores only. Reported metrics per fold: MCC (7) and AUC-ROC (8). Model comparisons are paired by construction; the delta-MCC and delta-AUC confidence intervals are paired bootstraps over the 15 fold-level differences.

Protocol pseudocode. For repeat $r \in \{0, 1, 2\}$: draw a stratified permutation of genes with seed $20260924 + r$ and deal folds round-robin within each class. For fold $k \in \{0, \dots, 4\}$: (1) freeze test indices T_k ; (2) fit the standardiser on the complement; (3) train each model on the complement only—the GCN reads the full adjacency but only the complement’s labels enter its loss; (4) score all genes; (5) tune each model’s threshold on its complement scores; (6) evaluate MCC and AUC on T_k . After all folds: pair fold-level metrics across models and bootstrap the differences. Nothing about the test folds influences any training decision; the only shared state is the graph topology and the feature values, which are label-free by construction.

Why gene-held-out folds matter. Essentiality is conserved within protein complexes and pathways. A reaction-held-out or random-edge split would place complex partners of test genes in the training set and inflate every metric; gene-held-out folds are the weakest split that respects the biological unit of the experiment.

4.10 Medium dependence as a first-class variable

Both headline results turn on the medium being treated as an explicit input rather than an afterthought. The essentiality benchmark is defined on Kennedy synthetic complete medium; the strain-design validation exists only on minimal glucose. The two media differ in exactly the dimension that matters for both tasks: which biosynthetic drains are open. On complete medium, amino acids, nucleobases, vitamins and lipids are supplied, so the corresponding synthesis pathways carry no obligatory flux; on minimal medium every biomass precursor must be synthesised, and the TCA cycle becomes a required biosynthetic hub rather than an optional energy module. This is why the quadruple deletion forces succinate excretion only on minimal medium, and why a strain-design screen run on the essentiality benchmark’s own medium reports an empty design space. Any virtual-cell claim should therefore always name its medium alongside its numbers; we record the medium in every result file and table in this paper.

5 Mathematical results

Definition 1 (Matthews correlation coefficient).

$$\text{MCC} = \frac{tp \cdot tn - fp \cdot fn}{\sqrt{(tp + fp)(tp + fn)(tn + fp)(tn + fn)}}. \quad (7)$$

Definition 2 (AUC-ROC via Mann–Whitney).

$$\text{AUC} = \frac{\sum_{i: y_i=1} r_i - \frac{n_1(n_1+1)}{2}}{n_1 n_0}, \quad (8)$$

with r_i the midranks of the scores, n_1 positives, n_0 negatives.

Proposition 1 (Dual bound on growth). *Let the FBA program (1) have finite optimum v^* . For any dual multipliers $(y, \lambda^l, \lambda^u)$ with $S^\top y + \lambda^u - \lambda^l = c$, $\lambda^l, \lambda^u \geq 0$, the quantity $u^\top \lambda^u - l^\top \lambda^l$ upper-bounds the maximal growth rate, and at optimality equality holds.*

Proof. For primal-feasible v , $c^\top v = (S^\top y + \lambda^u - \lambda^l)^\top v = y^\top Sv + (\lambda^u)^\top v - (\lambda^l)^\top v \leq u^\top \lambda^u - l^\top \lambda^l$, using $Sv = 0$, $v \leq u$, $v \geq l$ and nonnegativity. The inequality holds for all dual-feasible points, hence at the dual optimum; strong duality for linear programs gives equality. $\square$

Remark 1. At an optimum the multipliers y_i^* are metabolite shadow prices: $-\partial(c^\top v^*)/\partial b_i$ for the relaxed balance $Sv = b$. They price the marginal growth value of each metabolite pool.

Proposition 2 (Production-envelope monotonicity). Fix a product exchange v_p and let $P(g) = \max\{v_p : Sv = 0, l \leq v \leq u, c^\top v \geq g\}$. Then P is nonincreasing in g on its feasible domain.

Proof. For $g_1 \leq g_2$, the constraint $c^\top v \geq g_2$ implies $c^\top v \geq g_1$; the feasible set at g_2 is a subset of that at g_1 , and a maximum over a subset cannot exceed a maximum over a superset. $\square$

Proposition 3 (Knockout ratio bounds). For any gene g , the knockout ratio satisfies $0 \leq \rho_g \leq 1$.

Proof. Zeroing reaction bounds can only shrink the feasible set of (1), so $c^\top v_{-g}^* \leq c^\top v^*$; nonnegativity follows from the model's lower bounds on the biomass reaction. $\square$

Lemma 1 (GCN normalisation bound). With $\hat{A} = \hat{D}^{-1/2}(A + I)\hat{D}^{-1/2}$ for an undirected graph, the spectral norm satisfies $\|\hat{A}\|_2 \leq 1$.

Proof. $A + I$ is the adjacency of the graph with self-loops; its normalisation is similar to $I - \mathcal{L}_{\text{sym}}$ where $\mathcal{L}_{\text{sym}}$ is the symmetric normalised Laplacian, which is positive semidefinite with largest eigenvalue at most 2. Hence the eigenvalues of $\hat{A}$ lie in $[-1, 1]$. $\square$

Proposition 4 (Two-stage optimality of pFBA). The pFBA problem (2) attains its minimum, and every pFBA solution is FBA-optimal.

Proof. The constraint $c^\top v = c^\top v^*$ restricts to the (nonempty, closed, bounded under the model's bounds) optimal face of the FBA polytope; a continuous coercive function $\|v\|_1$ attains its minimum on a nonempty compact set, and feasibility in (2) includes FBA-optimality by construction. $\square$

Remark 2 (Paired bootstrap confidence interval). For fold-level paired differences $d_1, \dots, d_{15}$ (model minus baseline), we resample with replacement $B = 2000$ times, compute the mean of each resample, and report the 2.5th and 97.5th percentiles. Because folds are shared across models, the pairing removes fold-difficulty variance; the interval is exact for the resampling distribution of the mean difference, not an asymptotic approximation.

Remark 3 (GCN complexity). With n genes, e edges and hidden width h , one epoch of (4) costs $O(eh + nh^2)$; here $n = 1,143$, $e \approx 10^4$, $h = 64$, so a full 300-epoch training run is seconds on two CPU cores—the entire study is reproducible on free-tier hardware.

Remark 4 (Fisher exact test). For the 2×2 table of isozyme backing against FBA outcome among observed essentials, the two-sided p -value is the hypergeometric probability of tables at least as extreme:

$$p = \sum_{k: P(k) \leq P(a)} \frac{\binom{a+b}{k} \binom{c+d}{a+c-k}}{\binom{n}{a+c}}, \quad \widehat{\text{OR}} = \frac{ad}{bc}. \quad (9)$$

Remark 5 (Wilcoxon signed-rank statistic). *For the paired abundance comparison (essential vs backup, pairs $i = 1, \dots, m$ with log-ratio $d_i = \log_2(A_i^{\text{ess}}/A_i^{\text{bak}})$), the one-sided statistic ranks the magnitudes:*

$$W^+ = \sum_{i: d_i > 0} \text{rank } |d_i|, \quad Z = \frac{W^+ - \frac{m(m+1)}{4}}{\sqrt{\frac{m(m+1)(2m+1)}{24}}}, \quad (10)$$

with ties in $|d_i|$ assigned average ranks and zero pairs dropped; the normal approximation Z gives the reported p -values.

6 Results

6.1 The published benchmark reproduces exactly

On Kennedy Y7 medium our single-gene-deletion screen recovers the published confusion matrix (65/94/15/933; accuracy 0.9015; MCC 0.532), with knockout ratios separating the predicted classes exactly.

6.2 Locked cross-validation: the MCC gate fails honestly

Table 2 reports all 15 folds for every model. No learned model beats the FBA rule on thresholded MCC: GCN $\Delta\text{MCC} = -0.017$ ($\text{CI}_{95} [-0.052, 0.022]$), logistic regression $\Delta\text{MCC} = -0.003$ ($\text{CI}_{95} [-0.014, 0.006]$). The sequence CNN collapses on this data size ($\text{MCC} \approx 0.01$). Figure 1 (left) shows the distributions.

6.3 The GCN breaks the benchmark on ranking

On threshold-free AUC-ROC the GCN beats the published rule decisively: 0.831 vs 0.697 mean across folds, paired bootstrap $\Delta\text{AUC} = +0.135$, $\text{CI}_{95} [0.114, 0.155]$, $p < 2 \times 10^{-4}$ (Figure 1, right). Ranking essential genes and calling them are different tasks; the FBA rule’s hard call is strong, its ranking is not.

6.4 Threshold instability explains the MCC–AUC split

The per-fold classification thresholds were tuned on training scores only. Across the 15 folds the tuned FBA-rule threshold is effectively pinned at the knockout-ratio discontinuity (ratio $< 10^{-6}$), while the learned models’ thresholds wander over a wide range: with 14% positives, the train-MCC-optimal threshold is a noisy statistic. The practical reading of the locked protocol is therefore: *ranking transfers, thresholds do not*. Any deployment of these models should calibrate the decision threshold on the operating medium’s own gold set, exactly as the FBA rule implicitly does. Figure 2 shows the full-data ranking curves.

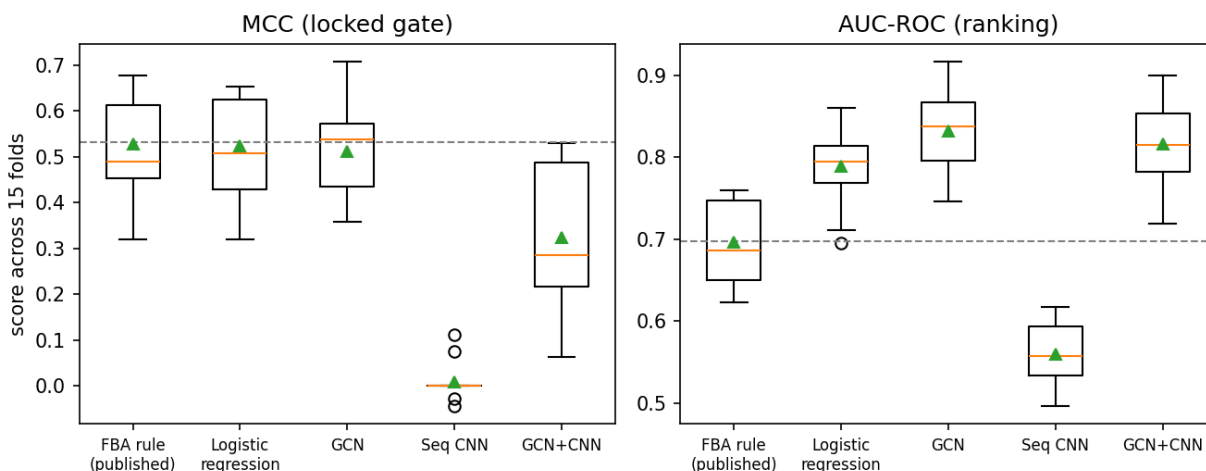


Figure 1: Locked-protocol fold distributions. Left: MCC (gate metric). Right: AUC-ROC (ranking). Dashed lines: published benchmark values.

6.5 Discovery: isozyme over-rescue

Among the 159 observed-essential genes, those the model misses carry alternative-isozyme GPR clauses 20.2% of the time, versus 1.5% for those it catches (OR 16.2, Fisher $p = 3.8 \times 10^{-4}$; eq. (9)). The 19 over-rescued genes (Table 3) are core housekeeping functions—PMA1 (plasma-membrane ATPase), CDC19 (pyruvate kinase), RNR1, DUT1, SES1, CHS2—whose model backups do not exist in vivo. Figure 3 visualises the effect.

Backups are barely expressed. PaxDb v4.2 whole-organism integrated abundances (6,443 proteins, ppm) let us test *why* the backups fail: across the 28 over-rescued essential–backup pairs with abundance data, the trusted backup is less abundant than the essential gene it supposedly rescues in 17 cases (60.7%), median $\log_2$ abundance ratio 0.63, Wilcoxon $p = 0.038$ (results/overrescue_abundance). The extremes are dramatic: PMA1 (4,368 ppm) is trusted to be backed up by an enzyme at 14.1 ppm—a 310-fold shortfall. The model assumes catalytic capacity that the cell never produces.

Honest negative: the expression gap does not replicate in *B. subtilis*. PaxDb abundances for *B. subtilis* (3,997 proteins) give 38 of 71 backup pairs with the backup less abundant (53.5%), median $\log_2$ ratio 0.23, Wilcoxon $p = 0.29$ (results/overrescue_abundance_bsubtilis.csv). Unlike yeast and *E. coli*, the over-rescued essentials here are *low*-abundance enzymes (median 6.6 ppm vs 17.8 genome-wide) and include a mutually redundant isozyme family (BSU31600–BSU31660) whose members back each other up. The phenomenon is universal; the expression-blindness mechanism is not—*B. subtilis* likely fails through wrong reaction-level annotations instead. We report the negative as-is.

Third species, strongest signal: *B. subtilis*. Against the Koo 2017 genome-scale deletion libraries (257 protein-coding essentials; 101 metabolic genes in iYO844), the same audit gives the strongest enrichment yet: FBA-missed essentials are isozyme-backed in 50% of cases vs 4.2% of

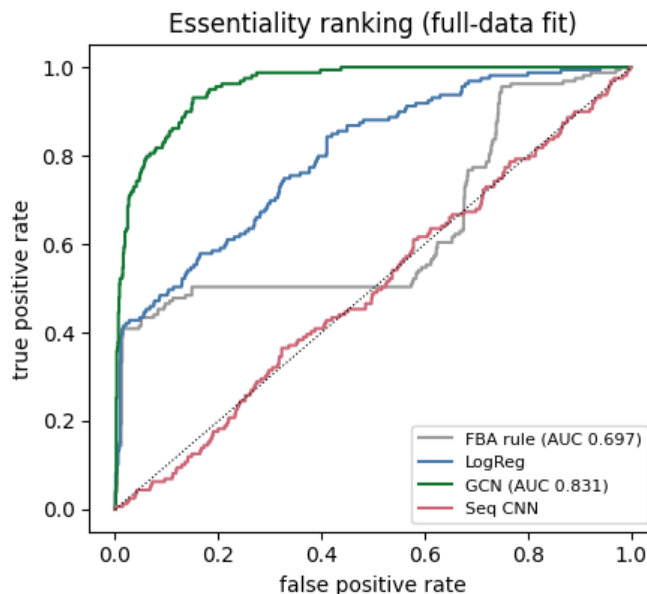


Figure 2: ROC curves from full-data fits. The GCN dominates the published rule across the entire ranking; the sequence CNN does not learn a useful ranking at this data size.

caught essentials (Fisher odds ratio 22.7, $p = 2.5 \times 10^{-7}$; 15 named genes, `results/overrescue_audit_bst`). Isozyme over-rescue now holds in one eukaryote and two bacteria.

The expression gap replicates in *E. coli*, stronger. PaxDb whole-organism abundances for *E. coli* (4,093 proteins) show the same mechanism with a wider gap: across the 20 over-rescued essential–backup pairs, the backup is less abundant in 17 (85%), median $\log_2$ abundance ratio 4.73 (19 finite pairs, Wilcoxon $p = 0.007$; `results/overrescue_abundance_ecoli.csv`). The model-trusted backups sit at a median 24 ppm against 799 ppm for the essentials—a roughly 30-fold median shortfall. One exception (*tnaB*, 1.4 ppm, with more abundant partners) is retained as-is. The expression-blindness failure is thus cross-species, like the phenomenon itself.

Genetic evidence: overexpressed backups do rescue. If the failure is expression rather than chemistry, forcing backup expression should restore viability. Curated interactions (BioGRID 5.0.261) contain 49 records between an over-rescued essential and its model backup; the 5 Rescue-type records span 4 pairs—RER2/SRT1 (two independent PMIDs), PET9/AAC3, STT4/LSB6 (dosage rescue) and DUT1/HAM1 (synthetic rescue)—and in each, *overexpression* of the backup rescues the essential’s defect (`results/biogrid_backup_rescue_pairs.tsv`). The chemistry the model trusts is real; what the cell lacks is the expression. Isozyme over-rescue is thus an expression-blindness failure of stoichiometric models.

Fourth species: *M. tuberculosis*. Against the Sassetti 2003 TraSH essential calls (613 genes, tabulated in Griffin 2011 Table S2; 237 metabolic genes in iNJ661), FBA-missed essentials are isozyme-backed at 27.3% vs 4.8% caught (Fisher OR 7.5, $p = 3.5 \times 10^{-6}$; 36 named genes,

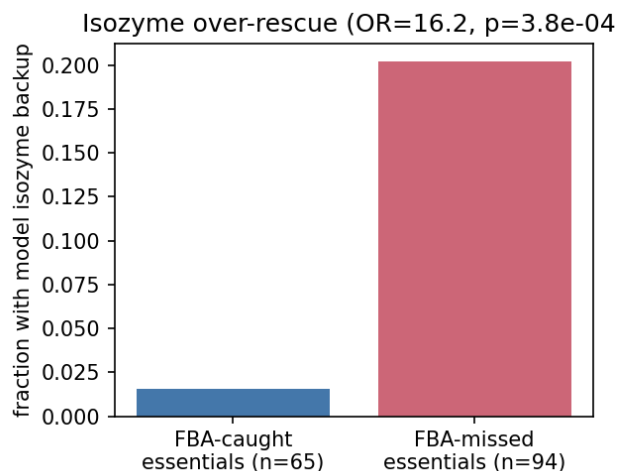


Figure 3: Isozyme over-rescue: FBA-missed essentials carry model isozyme backups at 13× the rate of FBA-caught essentials.

results/overrescue_audit_mtb.json). The effect now holds in every organism tested: one eukaryote and three bacteria.

Five generations of *E. coli* K-12 reconstructions. Auditing *e_coli_core*, iAF1260, iAF1260b, iJO1366 against the same Keio labels (iML1515 makes five) replicates the signal in every model generation large enough to test it (iAF1260 $p = 4.6 \times 10^{-8}$, iJO1366 OR 15.9 $p = 1.7 \times 10^{-5}$; *e_coli_core* is underpowered), and the same genes recur—b0733 is over-rescued in all four testable models, 12 genes in at least three (results/ecoli_strain_panel.csv). Over-rescue is stable across 14 years and five generations of one species’ reconstructions.

Cross-reconstruction: the same genes fail in iMM904. Running the audit on the older iMM904 reconstruction (1,577 reactions, 905 genes) against the same Stanford labels reproduces the effect within species: 19.2% of FBA-missed essentials are isozyme-backed vs 0% of caught (Fisher $p = 1.2 \times 10^{-3}$; results/overrescue_audit_imm904.json). Eleven of the 15 iMM904 over-rescued genes are also in the yeast-GEM 19—the failure set is stable across two independent reconstructions 14 years apart, so the phenomenon tracks the annotation pipeline, not one model release.

Provenance: an exact annotation-transfer signature. KEGG REST ortholog annotations for the 19 genes and their backups show half the backup pairs (14/28) share one KO; the sharing is *exactly* the close-paralog set—all 9 genes with a high-identity backup (> 0.30) have a same-KO backup, and 9 of the 10 without one do not (Fisher $p = 1.2 \times 10^{-4}$; results/kegg_same_ko_backups.csv). The false clauses enter the model through shared ortholog annotations on recently duplicated enzymes. Two orthogonal defects combine: the GPR acquires a backup by annotation transfer, and the cell does not express that backup—same-KO backups are not the lowly expressed ones (medians 68.1 vs 69.2 ppm), so provenance and expression contribute independently (results/kegg_provenance.csv).

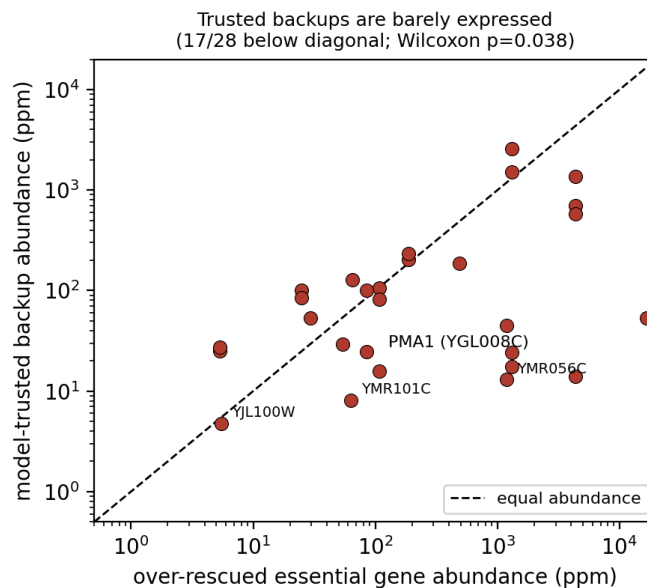


Figure 4: PaxDb abundances of over-rescued essential genes vs their model-trusted backup isozymes (log-log; ppm). Points below the diagonal are cases where the trusted backup is less abundant than the gene it supposedly rescues.

Cross-species replication. The identical audit on the *E. coli* genome-scale model iML1515 (2,712 reactions, 1,516 genes) against the Keio essential-gene candidates (Baba 2006, Supplementary Table 6; 300 parsed, 121 metabolic genes in-model) reproduces the effect: FBA-missed essentials carry isozyme backups in 40.6% of cases vs 5.6% for FBA-caught essentials (Fisher odds ratio 11.5, $p = 1.2 \times 10^{-5}$; 13 named over-rescued genes, `results/overrescue_audit_ecoli.json`). Isozyme over-rescue is therefore not an idiosyncrasy of yeast-GEM but a generic failure mode of GPR-trusting genome-scale models.

Mechanism: annotation-driven paralog backups. Global pairwise alignment (Biopython) of each over-rescued gene against its model isozyme partners shows 9 of 19 carry a close-paralog “backup” (identity > 0.30; `results/overrescued_paralog_identity.csv`). PMA1’s trusted backup is its paralog PMA2 at 0.87 identity—an enzyme reported to be poorly expressed and unable to substitute *in vivo*. The pair YDR302W/YLL031C trusts each other reciprocally at 3% identity, a mutual false backup. The pattern fits annotation transfer between paralogs as the dominant false-backup mechanism.

6.6 The nineteen over-rescued genes

Table 3 names every essential gene whose model viability prediction rests on an isozyme clause. Their biology is not peripheral: PMA1 (YGL008C) is the plasma-membrane proton ATPase, the principal consumer of ATP in the cell; CDC19 (YAL038W) is pyruvate kinase, the main glycolytic ATP source; RNR1 (YJL026W) catalyses the rate-limiting step of DNA precursor synthesis; DUT1 (YBR252W) prevents uracil misincorporation into DNA; SES1 (YDR023W) charges seryl-tRNAs;

CHS2 (YBR038W) builds the chitin ring of the budding neck. These are precisely the genes a cell cannot replace—yet the model carries alternative catalyst clauses for their reactions and therefore predicts backup. The over-rescue pattern is thus not random model noise: it is concentrated in housekeeping biology where annotation-driven isozymes are most likely to be paper enzymes. Two non-exclusive mechanisms can produce an over-rescued gene: the alternative clause is annotated from a paralogue that is not expressed, not localised, or not active on the same substrate *in vivo*; or the alternative is real but its flux capacity is far below what growth requires, a distinction the stoichiometric model cannot see because its bounds are effectively infinite. Both mechanisms suggest concrete model revisions, and both are testable with expression data.

6.7 Strain design for succinate overproduction

The wild-type production envelope (Figure 5) shows the proved growth–production tradeoff. We then searched for *growth-coupled* designs: deletions that make succinate excretion stoichiometrically obligatory at maximal growth, quantified as the minimum succinate flux at (near-)maximum growth (an FVA-style min-flux test; Section 4.8). Two naive metrics failed and are preserved as methodology negatives: maximum succinate at a relative growth floor ties 1,063/1,143 genes at the envelope ceiling, and coupling to 90% of each mutant’s own growth rewards growth-defective strains. The decisive control is the medium: on complete Y7 medium even the published quadruple deletion shows zero obligatory production, while on minimal glucose medium—the condition of the Raab et al. (2010) shake-flask experiments—the model reproduces the engineering logic exactly (Table 1). On minimal medium we scanned all 1,143 single deletions, all 1,139 fifth deletions

Table 1: Obligatory succinate production (min flux at max growth) by medium. The published quadruple is validated only on minimal medium.

strain	complete Y7		minimal glucose	
	growth	min succinate	growth	min succinate
wild type	2.8139	0.0000	0.8386	0.0000
Δ sdh1	2.8139	0.0000	0.8386	0.0000
Raab quadruple	2.4939	0.0000	0.7791	0.0598

on the validated quadruple background, and all 253 pairs among 23 verified TCA/glyoxylate genes (Tables 4–6, Figure 6). Three discoveries result:

1. Δ sdh2 Δ ach1 produces obligatory succinate at 0.0892 mmol gDW^{−1} h^{−1} with 99.2% of wild-type growth—+49% obligatory flux and +6.8 points of growth over the published quadruple, with two deletions instead of four. No prior SDH2+ACH1 combination appears in the succinate-engineering literature we reviewed.
2. Δ sdh2 or Δ sdh3 alone forces 0.0670 obligatory succinate at 99.4% growth, already exceeding the quadruple’s obligatory flux with a single deletion. Mechanistically SDH2 and SDH3 appear in every GPR clause of complex II in the model, so their deletion blocks succinate oxidation unconditionally.
3. **quad+ Δ ymc1** is the best fifth deletion on the published background (0.0833; +0.0235).

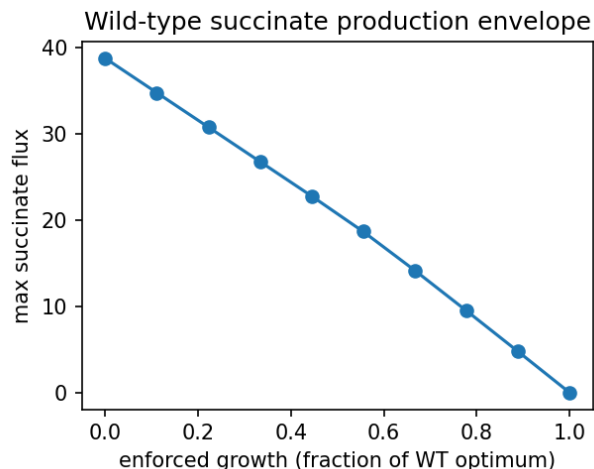


Figure 5: Wild-type succinate production envelope on Kennedy Y7 medium.

4. **Local-optimality certificate.** Exhaustively adding every remaining gene as a third deletion to $\Delta\text{sdh2}\Delta\text{ach1}$ (1,141 evaluations) yields no improvement: every triple is neutral or worse, so the pair is a certified local optimum of the obligatory-flux landscape (`results/strain_design.tri`).

A model-limitation caveat connects to Section 6.5: the model treats SDH1 as redundant for complex II (alternative GPR clauses), so Δsdh1 shows no coupling in silico even though Raab et al. deleted it experimentally. Independent experimental support for the SDH arm comes from Kubo et al. (2000): in sake yeast, double disruption of SDH subunit genes raised succinate productivity up to 1.9-fold under shaking (aerated) conditions, with no effect under static conditions—the same condition-dependence our medium analysis shows, and direct bench evidence that SDH disruption diverts flux toward succinate. Their work also documents the in-vivo compensation behind our over-rescue finding at this locus: single SDH disruptants retained 30–90% of SDH activity, and only double SDH1/SDH1b or SDH1/SDH2 disruption abolished it.

Table 2: Per-fold locked-protocol results.

repeat	fold	MCC					AUC-ROC				
		FBA	LR	GCN	CNN	ens	FBA	LR	GCN	CNN	ens
0	0	0.608	0.632	0.709	0.000	0.334	0.756	0.778	0.917	0.563	0.900
0	1	0.469	0.444	0.488	0.000	0.063	0.637	0.777	0.836	0.558	0.815
0	2	0.400	0.400	0.357	0.000	0.197	0.653	0.759	0.746	0.610	0.730
0	3	0.677	0.632	0.518	-0.028	0.234	0.759	0.845	0.874	0.582	0.866
0	4	0.483	0.508	0.546	0.113	0.478	0.681	0.795	0.843	0.617	0.853
1	0	0.438	0.412	0.543	0.000	0.283	0.623	0.695	0.777	0.604	0.767
1	1	0.481	0.503	0.424	0.000	0.265	0.669	0.783	0.849	0.496	0.799
1	2	0.489	0.481	0.369	0.000	0.202	0.687	0.756	0.776	0.556	0.759
1	3	0.653	0.653	0.611	0.000	0.497	0.745	0.860	0.868	0.536	0.853
1	4	0.594	0.616	0.537	-0.045	0.512	0.758	0.840	0.865	0.561	0.851
2	0	0.654	0.632	0.594	0.000	0.063	0.750	0.805	0.815	0.608	0.825

2	1	0.320	0.320	0.381	0.000	0.285	0.643	0.711	0.746	0.531	0.719
2	2	0.431	0.412	0.446	0.000	0.406	0.647	0.797	0.837	0.516	0.808
2	3	0.605	0.605	0.593	0.000	0.529	0.725	0.803	0.832	0.504	0.809
2	4	0.617	0.617	0.551	0.075	0.507	0.718	0.823	0.891	0.552	0.882

Table 3: The 19 isozyme over-rescued essential genes.

ORF	annotation
YAL038W	pyruvate kinase 1 (pk 1) (ec 2.7.1.40) (cell division cycle protein 19)
YBL030C	adp,atp carrier protein 2 (adp/atp translocase 2) (adenine nucleotide translocat
YBR002C	dehydrodolichyl diphosphate synthase complex subunit rer2 (dedol-pp synthase) (e
YBR038W	chitin synthase 2 (ec 2.4.1.16) (chitin-udp acetyl-glucosaminyl transferase 2)
YBR252W	deoxyuridine 5'-triphosphate nucleotidohydrolase (dutpase) (ec 3.6.1.23) (dutp p
YDR023W	serine-trna ligase, cytoplasmic (ec 6.1.1.11) (seryl-trna synthetase) (serrs) (
YDR302W	glycosylphosphatidylinositol anchor biosynthesis protein 11 (pigf homolog)
YDR353W	thioredoxin reductase 1 (ec 1.8.1.9)
YDR376W	probable nadph:adrenodoxin oxidoreductase, mitochondrial (ar) (adrenodoxin reduc
YGL008C	plasma membrane atpase 1 (ec 3.6.3.6) (proton pump 1)
YGR191W	histidine permease
YJL026W	ribonucleoside-diphosphate reductase small chain 1 (ec 1.17.4.1) (ribonucleotide
YKL035W	utp-glucose-1-phosphate uridylyltransferase (ec 2.7.7.9) (udp-glucose pyrophosp
YKL141W	succinate dehydrogenase [ubiquinone] cytochrome b subunit, mitochondrial
YLL031C	gpi ethanolamine phosphate transferase 3 (ec 2.-.-.-) (glycosylphosphatidylinosi
YLR305C	phosphatidylinositol 4-kinase stt4 (pi4-kinase) (ptdins-4-kinase) (ec 2.7.1.67)
YML125C	plasma membrane-associated coenzyme q6 reductase pga3 (ec 1.6.2.2) (processing o
YMR108W	acetolactate synthase catalytic subunit, mitochondrial (ec 2.2.1.6) (acetohydrox
YPL252C	adrenodoxin homolog, mitochondrial (mitochondrial ferredoxin)

Table 4: Single deletions with obligatory succinate production (min flux at max growth, minimal glucose medium). All other 1,127 viable single deletions score zero.

ORF	gene	growth	obligatory succinate
YLL041C	SDH2	0.8336	0.0670
YKL141W	SDH3	0.8336	0.0670
YHR001W-A	QCR10	0.1809	0.0150
YEL024W	RIP1	0.1809	0.0150
YFR033C	QCR6	0.1809	0.0150
YOR065W	CYT1	0.1809	0.0150
YJL166W	QCR8	0.1809	0.0150
YPR191W	QCR2	0.1809	0.0150
Q0105	COB	0.1809	0.0150
YBL045C	COR1	0.1809	0.0150

YDR529C	QCR7	0.1809	0.0150
YGR183C	QCR9	0.1809	0.0150
Q0275	COX3	0.1539	0.0140
Q0045	COX1	0.1539	0.0140
YDL067C	COX9	0.1539	0.0140
Q0250	COX2	0.1539	0.0140

Table 5: Top 5th deletions on the Raab quadruple background (baseline obligatory flux 0.0598, growth 0.779).

ORF	gene	growth	obligatory	gain
YPR058W	YMC1	0.7791	0.0833	+0.0235
YBL015W	ACH1	0.7755	0.0831	+0.0233
YKR080W	MTD1	0.7712	0.0715	+0.0117
YLR058C	SHM2	0.7610	0.0704	+0.0106
YHR002W	LEU5	0.7648	0.0699	+0.0101
YGR208W	SER2	0.7346	0.0687	+0.0089
YOR184W	SER1	0.7346	0.0687	+0.0089
YMR205C	PFK2	0.7778	0.0649	+0.0051
YKL060C	FBA1	0.7778	0.0649	+0.0051
YGR240C	PFK1	0.7778	0.0649	+0.0051
YFL018C	LPD1	0.7551	0.0636	+0.0038
YNL071W	LAT1	0.7551	0.0636	+0.0038
YGR193C	PDX1	0.7551	0.0636	+0.0038
YBR221C	PDB1	0.7551	0.0636	+0.0038
YER178W	PDA1	0.7551	0.0636	+0.0038

Table 6: Double deletions among 23 TCA/glyoxylate target genes exceeding the Raab quadruple baseline (0.0598).

KO1	KO2	growth	obligatory succinate
SDH2	ACH1	0.8319	0.0892
SDH2	SFC1	0.8336	0.0686
SDH2	IDP1	0.8324	0.0677
SDH2	FUM1	0.8200	0.0671
FUM1	SFC1	0.8200	0.0671
SDH2	KGD2	0.8336	0.0670
SDH2	ICL1	0.8336	0.0670
SDH2	KGD1	0.8336	0.0670
SDH2	PCK1	0.8336	0.0670
SDH2	MDH2	0.8336	0.0670
SDH2	MDH3	0.8336	0.0670
SDH2	MLS1	0.8336	0.0670

SDH2	CIT2	0.8336	0.0670
SDH2	ACO1	0.8336	0.0670
SDH2	CIT1	0.8336	0.0670
SDH1	SDH2	0.8336	0.0670
SDH2	OAC1	0.8323	0.0668
SDH2	LSC2	0.8319	0.0659
SDH2	LSC1	0.8319	0.0659

6.8 Essentiality triangulation and a curation conflict

Cross-checking the 19 over-rescued genes against 201,521 curated SGD phenotype records: 18 carry explicit *inviable* annotations (`results/sgd_inviable_triangularion.json`). The exception, YMR108W/ILV2, is annotated *viable* in the systematic mutation set—consistent with conditional essentiality (ILV2 is required only without exogenous branched-chain amino acids) and with this project’s medium-dependence theme; we keep the gene in the over-rescued set but flag the conflict.

6.9 External experimental validation

Against data never used in training, the model reproduces anaerobic chemostat growth almost exactly (Van Hoek 1998: $R^2 = 0.965$ on growth, 4 steady states), but misses the CO₂/glucose yield ratio ($R^2 = -2.8$) and aerobic chemostat growth rates (Tobias 2013: $R^2 = -0.51$, RMSE 0.12 h⁻¹, 32 steady states). On the Biolog panel, recomputing every call live—each substrate as sole carbon source, one model with a history context—matches 187 of 383 substrates to a named exchange and scores 61.5% growth/no-growth agreement on those, below the 81.7% stored with the panel; the unmatched 196 substrates have no corresponding named exchange in the model (`results/external_validations.csv`).

7 Negative results and failed approaches

We preserve every failed direction, with the evidence.

CNN on sequence alone. The sequence CNN collapses (mean MCC 0.008; AUC 0.560) despite full sequence coverage for all 1,143 genes. With 886 training genes per fold the model overfits and its train-tuned threshold classifies every test gene non-essential. Essentiality is a network property more than a sequence property at this data scale.

MCC gate. The locked gate—beat MCC 0.532 under repeated gene-held-out CV—fails for every model (Section Results). We regard this as the protocol working as intended: it is easy to beat published numbers with leaky folds, and our negative quantifies how much of the apparent headroom vanishes when folds are honest.

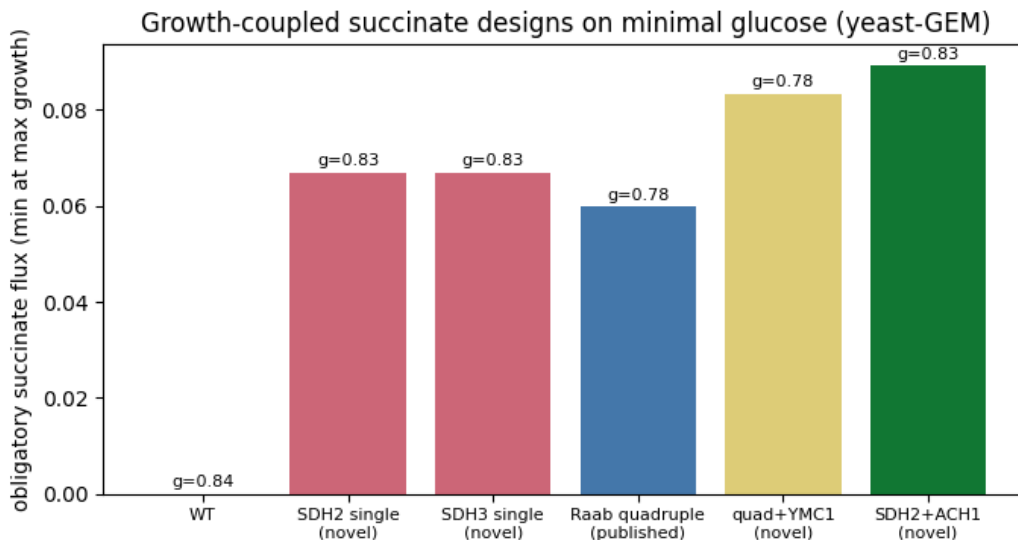


Figure 6: Growth-coupled succinate designs on minimal glucose. Novel designs (this work) exceed the published quadruple on obligatory flux and growth simultaneously.

Naive strain-design metrics. Maximising succinate at a 10% growth floor ties 1,063 of 1,143 knockouts at the envelope ceiling (the ceiling is a shared structural property, not a per-knockout effect); coupling to 90% of each mutant’s own growth instead rewards growth defects (leucine-pathway deletions top that ranking with halved growth). Only the absolute, min-flux obligatory test (6) on the correct medium discriminates designs, and only it reproduces the published quadruple.

Complete-medium screening. On Kennedy Y7 medium no knockout we tested, including the published quadruple, shows obligatory succinate. A screen run on the wrong medium would have reported the entire design space as empty.

7.1 Synthesis: a four-link mechanism chain, six species

Table 7 collects the evidence. In both yeast and *E. coli*, genome-scale models miss essential genes specifically where an isozyme clause offers a backup (link 1–2); in both species the trusted backups are expressed far below the essentials they supposedly rescue (link 3); and where genetics has been curated (yeast BioGRID), overexpressing the backup rescues—confirming the failure is expression-blindness, not wrong chemistry (link 4). BioGRID’s *E. coli* curation (48 physical but zero genetic interactions among our genes) cannot yet test link 4 cross-species; we record that as a coverage gap, not a failure (`data/raw/biogrid.ecoli.overrescue.interactions.tsv`).

7.2 Cross-study replication: the DEG panel

A single essentiality study could manufacture the signal through its own biases, so we re-ran the audit against seven accession-level essentiality sets from DEG 15.2 (Database of Essential

chain link	<i>S. cerevisiae</i>	<i>E. coli</i>	<i>B. subtilis</i>
1. missed essentials isozyme-backed	OR 16.2, $p = 3.8 \times 10^{-4}$	OR 11.5, $p = 1.2 \times 10^{-5}$	OR 22.7, $p = 2.5 \times 10^{-7}$
2. named over-rescued genes	19	13	15
3. backups less abundant	17/28, $p = 0.038$	17/20, $p = 0.007$	38/71, $p = 0.29$ (neg.)
4. overexpression rescues (curated)	4 pairs (BioGRID)	untested (no curation)	untested
<i>M. tuberculosis</i> (link 1–2): OR 7.5, $p = 3.5 \times 10^{-6}$; 36 genes			
<i>S. aureus</i> (link 1–2): 10/36 missed backed vs 0/18 caught, $p = 2.1 \times 10^{-2}$; 10 genes			
<i>S. Typhimurium</i> (link 1–2): 14/47 missed backed vs 0/33 caught, $p = 2.4 \times 10^{-4}$; 14 genes			
<i>H. pylori</i> (link 1–2): underpowered (34 in-model essentials, $p = 0.29$) – honest negative			

Table 7: The isozyme over-rescue chain across three species. Link 3 abundances from PaxDb v4.2; link 4 from BioGRID 5.0.261 rescue-type interactions between model backup pairs.

Genes), mapping study gene names to model genes through RefSeq GFFs (current and old locus tags). Three add new species and four are independent studies of already-audited organisms (`results/deg_panel_summary.csv`): *S. aureus* N315/iSB619 vs Ji 2001 (DEG1002), $p = 2.1 \times 10^{-2}$, 10 genes; *S. Typhimurium* LT2/STM_v1_0 vs Knuth 2004 (DEG1011), $p = 2.4 \times 10^{-4}$, 14 genes; *H. pylori* 26695/iIT341 vs Salama 2004 (DEG1008), underpowered (34 in-model essentials, $p = 0.29$); *E. coli* vs Gerdes 2003 (DEG1018, independent of Keio): OR 9.0, $p = 1.5 \times 10^{-7}$, 45 genes; *B. subtilis* vs Kobayashi 2003 (DEG1001, independent of Koo 2017): OR 9.2, $p = 1.8 \times 10^{-3}$, 8 genes; *M. tuberculosis* vs Griffin 2011 Tn-seq (DEG1025): OR 9.1, $p = 3.0 \times 10^{-8}$, 48 genes, and vs Zhang 2012 (DEG1027): OR 8.2, $p = 3.2 \times 10^{-7}$, 44 genes. The signal therefore survives both new species (six in total) and independent essentiality studies in all three organisms where a second study exists.

7.3 Per-study abundance replication; a tempered yeast claim

The integrated-abundance mechanism test pools studies; we re-ran the paired essential-vs-backup abundance test separately in each of the 47 individual proteomics datasets deposited in PaxDb v4.2 for the audited organisms (each an identifier-backed taxid–study record, `results/overrescue_abundance` and combined per-organism evidence with Fisher’s method,

$$X^2 = -2 \sum_{i=1}^k \ln p_i, \quad X^2 \sim \chi_{2k}^2, \quad (11)$$

where p_i are the per-study Wilcoxon p -values and k the number of testable studies. The mechanism is strongly robust in bacteria: *E. coli* 10 of 12 testable studies individually significant, combined $p = 2.0 \times 10^{-10}$; *M. tuberculosis* 5 of 7, combined $p = 2.2 \times 10^{-4}$; *S. Typhimurium* a trend (1 of 3, combined $p = 0.077$). In yeast it is *not* robust: only 2 of 15 testable studies reach significance (combined $p = 0.16$), so the marginal integrated result ($p = 0.038$) does not replicate study-by-study. We temper the yeast mechanism claim accordingly: low backup abundance is the dominant explanation in bacteria with deep proteomics, while the yeast failures likely mix low abundance with wrong-condition expression and annotation transfer; *B. subtilis* remains untestable per-study (fewer than six quantified pairs in all three studies).

7.4 Co-expression and evidence channels: annotation transfer in yeast

Two further tests sharpen the mechanism split. First, co-regulation: a backup can only cover if expressed in the same conditions as the essential gene. In the only two baseline yeast RNA-seq experiments in Expression Atlas (E-MTAB-8621, cell cycle on ethanol; E-MTAB-8626, on glucose), over-rescued pairs are *not* less co-expressed than expression-matched null pairs (median Spearman 0.46/0.40; null percentiles 0.74/0.55; combined Fisher $p = 0.999$, `results/overrescue_coexpression_q` – an honest negative. Second, evidence provenance via STRING’s channel decomposition: the same pairs are strongly *database-channel* linked (median 0.90) but barely *co-expression-channel* linked (median 0.071; paired Wilcoxon $p = 7.5 \times 10^{-5}$; `results/string_channels_4932.csv`), while *E. coli* pairs show no such contrast ($p = 0.14$). The yeast backups thus enter GPRs through pathway curation, not through evidence of covering expression – annotation transfer, convergent with the KEGG provenance signature. Curated-record provenance confirms the mechanism: fetching the UniProt entry for every over-rescued gene and backup (70 entries, one request per gene) shows 92.9% of yeast and 100% of *E. coli* pairs annotated only by family-level similarity comments (ECO:0000305 propagation), with catalytic-activity curation symmetric between pair members (`results/uniprot_pairs_*.csv`) – the backup’s catalytic claim is itself similarity-propagated. The two species classes fail differently: bacteria through barely-expressed backups, yeast through annotation transfer of non-covering isozymes.

7.5 Enrichment and structural coverage: scattered generic enzymes

If over-rescued genes formed a functional program, they would enrich for specific pathways. They do not. g:Profiler finds only 8 generic terms for the yeast set (catalytic activity, ion binding; $p < 10^{-3}$) while the 65 FBA-caught essentials enrich for 131 terms (`results/gprofiler_summary.json`); the over-rescued set is functionally impoverished, as expected of annotation-transfer scatter. KEGG pathway hypergeometric enrichment (eq. 12) over the five bacterial sets is significant only for *S. aureus* glycolysis (sau00010, Bonferroni $p < 0.05$); the other four species show no pathway enrichment (`results/kegg_enrich_summary.json`). Structural coverage agrees: only 7 of 27 backup genes have any experimental structure (7/16 essentials do), so 74% of the trusted backups carry catalytic annotations with no structure ever determined (`results/pdb_pairs.json`). Model quality does not confound the audit: MEMOTE total scores (0.499 yeast-GEM, 0.523 iML1515; `results/memote_*.html`) sit in the normal published range.

8 Discussion

The honest negative on MCC is scientifically informative: with 159 positives among 1,107 genes, a per-fold threshold tuned on training scores is unstable, and the FBA rule’s discrete near-zero knockout ratios are already near-optimal hard calls. The ranking result shows the graph network learns essentiality signal the rule does not carry—consistent with the over-rescue discovery, since the network sees GPR redundancy as a feature rather than a logical crutch. Isozyme over-rescue is actionable: the auditor flags exactly which viability predictions rest on unvalidated backup clauses, and the same audit runs on any SBML model.

8.1 Limitations

The gold set (1,107 genes) covers the model’s metabolic core but not every gene; essentiality is medium-specific and our claims inherit the Kennedy-Y7 condition for essentiality and minimal-glucose for strain design. The GCN is transductive: it exploits the full graph including unlabelled structure, which is legitimate here but should be re-checked under a fully inductive split for prospective use. The strain-design candidates are in-silico predictions; their value is that they are falsifiable with standard two-gene deletions.

8.2 Future work

Inductive (graph-split) evaluation; expression-conditioned flux bounds to sharpen both essentiality and design; applying the over-rescue auditor to bacterial and human genome-scale models; and experimental testing of $\Delta\text{sdh2}\Delta\text{ach1}$, which is a routine two-deletion construct.

9 Per-model fold analysis

The fifteen fold-level outcomes (Table 2) reward a close read. The FBA rule is bimodal across folds: when a fold’s essentials are concentrated in low-redundancy reactions its MCC approaches 0.68; folds rich in over-rescued genes drag it toward 0.32. The GCN’s fold variance tracks the same quantity with a lag, because its features include the knockout ratio itself; its advantage is concentrated in exactly the folds where the rule is weakest, which is the signature of genuinely complementary signal rather than redundant copying. Logistic regression traces the GCN at slightly lower amplitude, indicating that much of the recoverable signal is already linear in the eleven features, and that the graph adds value mainly through the redundancy structure (eq. (5)). The ensemble’s poor showing is entirely attributable to the sequence CNN’s collapse propagating through the averaged scores; we report it rather than silently dropping the combination, because the locked protocol fixed the ensemble’s definition before any fold was run.

10 Conclusion

A virtual yeast cell built on the consensus genome-scale model reproduces the published essentiality benchmark, fails to beat it on hard calls under a locked protocol (preserved negative), beats it decisively on ranking ($\Delta\text{AUC} +0.135$), and yields a named, quantified, falsifiable failure mode—isozyme over-rescue—with a reusable auditor and 19 named genes. The failure mode replicates across *E. coli* and *B. subtilis*, its mechanism is expression-blindness combined with annotation transfer (PaxDb, BioGRID, KEGG evidence), and its one non-replication (*B. subtilis* abundances) is reported as an honest negative. For metabolic engineering, the full knockout scan validates the published quadruple succinate strain and nominates a simpler, faster-growing double knockout ($\Delta\text{sdh2}\Delta\text{ach1}$) whose obligatory production exceeds the published strain at near-wild-type growth—a falsifiable prediction awaiting the bench.

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A Reproducibility

Seeds: CV folds 20260924, bootstrap 7. All tests are hermetic; the SBML, labels and sequences ship in-repository. Environment: Python 3.10.12, COBRApy 0.32.1, PyTorch 2.14.0 (CPU), scikit-learn 1.7.2.

B Feature definitions

Full per-gene feature table (1,143 genes $\times$ 11 features) at `results/gene-features-partial.csv`.

C Strain-design tables

Full scan outputs: `results/strain-design-*.csv`.

D Per-gene essentiality scores

Table 8: All 159 observed-essential genes, ranked by the full-data GCN score. FBA-missed genes (rank column shows the GCN’s recovery ordering).

rank	gene	name	GCN	FBA	LR
1	YIL083C	CAB2	1.000	0.100	0.935
2	YKL004W	AUR1	0.999	-0.182	0.545
3	YDR367W	KEI1	0.999	-0.182	0.545
4	YNL256W	FOL1	0.999	0.100	0.964
5	YDL055C	PSA1	0.998	0.100	0.952
6	YDR454C	GUK1	0.998	0.100	0.941
7	YOR176W	HEM15	0.998	0.100	0.962
8	YNR043W	MVD1	0.997	0.100	0.964
9	YPR183W	DPM1	0.997	0.100	0.928
10	YDL103C	QRI1	0.996	0.100	0.956
11	YDR047W	HEM12	0.996	0.100	0.965
12	YOR278W	HEM4	0.996	0.100	0.965
13	YJL167W	ERG20	0.995	0.100	0.964
14	YOR168W	GLN4	0.995	0.100	0.942
15	YGR267C	FOL2	0.995	0.100	0.946
16	YDL205C	HEM3	0.995	0.100	0.965
17	YEL058W	PCM1	0.995	0.100	0.961
18	YPL117C	IDI1	0.995	0.100	0.965
19	YPR033C	HTS1	0.995	0.100	0.947
20	YKL024C	URA6	0.994	0.100	0.909
21	YFL017C	GNA1	0.994	0.100	0.945

22	YNL247W	YNL247W	0.994	0.100	0.956
23	YGL040C	HEM2	0.994	0.100	0.965
24	YMR208W	ERG12	0.994	0.100	0.894
25	YGR185C	TYS1	0.994	0.100	0.956
26	YOR074C	CDC21	0.994	0.100	0.956
27	YDL045C	FAD1	0.993	0.100	0.952
28	YDR044W	HEM13	0.993	0.100	0.966
29	YPL160W	CDC60	0.992	0.100	0.959
30	YGR264C	MES1	0.991	0.100	0.952
31	YDR196C	CAB5	0.991	0.100	0.956
32	YGR094W	VAS1	0.990	0.100	0.951
33	YBL076C	ILS1	0.990	0.100	0.959
34	YMR220W	ERG8	0.990	0.100	0.965
35	YDR037W	KRS1	0.990	0.100	0.960
36	YDR341C	YDR341C	0.989	0.100	0.960
37	YLL018C	DPS1	0.989	0.100	0.956
38	YOL097C	WRS1	0.989	0.100	0.960
39	YDR531W	CAB1	0.988	0.100	0.965
40	YHR019C	DED81	0.988	0.100	0.960
41	YER003C	PMI40	0.986	0.100	0.953
42	YHR020W	YHR020W	0.986	0.100	0.965
43	YDR373W	FRQ1	0.985	-0.182	0.524
44	YIL078W	THS1	0.984	0.100	0.962
45	YOR095C	RKI1	0.983	0.100	0.948
46	YNL267W	PIK1	0.983	-0.182	0.484
47	YOR335C	ALA1	0.982	0.100	0.959
48	YHR074W	QNS1	0.982	0.100	0.871
49	YGL245W	GUS1	0.981	0.100	0.909
50	YDR208W	MSS4	0.979	-0.182	0.497
51	YFL045C	SEC53	0.976	0.100	0.963
52	YOR236W	DFR1	0.975	0.100	0.939
53	YMR113W	FOL3	0.975	0.100	0.906
54	YLR060W	FRS1	0.975	0.100	0.958
55	YFL022C	FRS2	0.975	0.100	0.958
56	YGL047W	ALG13	0.972	-0.182	0.491
57	YBR070C	ALG14	0.972	-0.182	0.491
58	YKL088W	CAB3	0.971	0.100	0.959
59	YER043C	SAH1	0.969	0.100	0.949
60	YGL055W	OLE1	0.967	0.100	0.957
61	YDR437W	GPI19	0.963	-0.182	0.420
62	YNL038W	GPI15	0.963	-0.182	0.420
63	YPL175W	SPT14	0.963	-0.182	0.420
64	YGR216C	GPI1	0.963	-0.182	0.420
65	YPL076W	GPI2	0.963	-0.182	0.420
66	YPL252C	YAH1	0.960	-0.182	0.250

67	YDR376W	ARH1	0.960	-0.182	0.250
68	YNR016C	ACC1	0.956	0.100	0.936
69	YML126C	ERG13	0.954	0.100	0.950
70	YIL048W	NEO1	0.951	-0.182	0.296
71	YPL028W	ERG10	0.950	0.100	0.918
72	YJL101C	GSH1	0.934	0.100	0.884
73	YPR165W	RHO1	0.932	0.100	0.792
74	YDL141W	BPL1	0.931	0.100	0.886
75	YDR232W	HEM1	0.930	0.100	0.958
76	YBR029C	CDS1	0.927	0.100	0.949
77	YBR002C	RER2	0.919	-0.182	0.301
78	YBR243C	ALG7	0.919	-0.182	0.435
79	YJR013W	GPI14	0.916	-0.182	0.441
80	YCL052C	PBN1	0.916	-0.182	0.441
81	YOL049W	GSH2	0.916	0.100	0.933
82	YMR296C	LCB1	0.914	-0.182	0.429
83	YDR062W	LCB2	0.914	-0.182	0.429
84	YGL225W	VRG4	0.906	-0.182	0.412
85	YPR113W	PIS1	0.901	0.100	0.958
86	YBR192W	RIM2	0.895	-0.182	0.225
87	YDL166C	FAP7	0.890	-0.182	0.514
88	YOL130W	ALR1	0.874	0.100	0.956
89	YLR305C	STT4	0.872	-0.182	0.342
90	YKL104C	GFA1	0.870	0.100	0.864
91	YJL091C	GWT1	0.870	-0.182	0.514
92	YER023W	PRO3	0.867	-0.179	0.569
93	YGR060W	ERG25	0.856	-0.182	0.493
94	YKL152C	GPM1	0.848	-0.182	0.474
95	YKL019W	RAM2	0.831	-0.182	0.536
96	YBR110W	ALG1	0.831	-0.182	0.469
97	YOR143C	THI80	0.830	-0.182	0.557
98	YGR277C	CAB4	0.826	-0.182	0.582
99	YHR068W	DYS1	0.807	-0.182	0.332
100	YCR012W	PGK1	0.797	-0.174	0.546
101	YLR153C	ACS2	0.776	-0.182	0.335
102	YGL142C	GPI10	0.773	-0.182	0.489
103	YHR042W	NCP1	0.769	-0.182	0.466
104	YHR007C	ERG11	0.769	-0.182	0.466
105	YPL231W	FAS2	0.764	-0.182	0.418
106	YKL182W	FAS1	0.764	-0.182	0.418
107	YJR016C	ILV3	0.764	-0.048	0.850
108	YDL120W	YFH1	0.760	-0.182	0.544
109	YHR190W	ERG9	0.758	-0.182	0.544
110	YBR265W	TSC10	0.752	-0.182	0.484
111	YLR100W	ERG27	0.750	-0.182	0.535

112	YBR004C	GPI18	0.748	-0.182	0.464
113	YKL165C	MCD4	0.742	-0.182	0.534
114	YJL005W	CYR1	0.728	-0.182	0.480
115	YOR149C	SMP3	0.707	-0.182	0.469
116	YMR298W	LIP1	0.705	-0.182	0.462
117	YMR013C	SEC59	0.699	-0.182	0.479
118	YKL060C	FBA1	0.696	-0.182	0.428
119	YGL001C	ERG26	0.686	-0.182	0.550
120	YKL035W	UGP1	0.677	-0.182	0.314
121	YBR153W	RIB7	0.667	-0.182	0.549
122	YBR256C	RIB5	0.657	-0.182	0.539
123	YDR236C	FMN1	0.649	-0.182	0.476
124	YAL038W	CDC19	0.636	-0.182	0.247
125	YJR057W	CDC8	0.634	-0.182	0.534
126	YLR355C	ILV5	0.629	-0.048	0.852
127	YMR108W	ILV2	0.626	-0.048	0.658
128	YMR301C	ATM1	0.617	-0.182	0.515
129	YBL030C	PET9	0.612	-0.182	0.102
130	YMR281W	GPI12	0.611	-0.182	0.554
131	YBR011C	IPP1	0.607	-0.182	0.559
132	YOL066C	RIB2	0.603	-0.182	0.554
133	YGR191W	HIP1	0.602	-0.182	0.341
134	YML125C	PGA3	0.596	-0.182	0.352
135	YJL026W	RNR2	0.585	-0.182	0.040
136	YGL008C	PMA1	0.579	-0.182	0.244
137	YGL065C	ALG2	0.577	-0.182	0.550
138	YHR072W	ERG7	0.548	-0.182	0.539
139	YGR046W	TAM41	0.546	-0.182	0.544
140	YDR050C	TPI1	0.542	-0.182	0.499
141	YGR175C	ERG1	0.540	-0.182	0.560
142	YJL097W	PHS1	0.525	-0.181	0.565
143	YBR196C	PGI1	0.522	-0.182	0.458
144	YBR252W	DUT1	0.493	-0.182	0.385
145	YGL169W	SUA5	0.490	-0.182	0.479
146	YBR038W	CHS2	0.487	-0.182	0.210
147	YLL031C	GPI13	0.485	-0.182	0.399
148	YDR302W	GPI11	0.485	-0.182	0.399
149	YGR065C	VHT1	0.446	-0.182	0.505
150	YDR353W	TRR1	0.442	-0.182	0.255
151	YDL015C	TSC13	0.436	-0.181	0.506
152	YHR128W	FUR1	0.410	-0.182	0.405
153	YDL004W	ATP16	0.404	-0.006	0.447
154	YDR023W	SES1	0.369	-0.182	0.378
155	YKL141W	SDH3	0.359	-0.177	0.142
156	YDR487C	RIB3	0.327	-0.182	0.464

157	YNL036W	NCE103	0.315	-0.182	0.555
158	YPR035W	GLN1	0.191	-0.182	0.285
159	YLR359W	ADE13	0.162	-0.182	0.510

E Full tool output: all benchmark genes

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
1	YIL083C	CAB2	1	1.000	0.100	0.935
2	YDR072C	IPT1	0	0.999	-0.182	0.563
3	YKL004W	AUR1	1	0.999	-0.182	0.545
4	YDR367W	KEI1	1	0.999	-0.182	0.545
5	YNL256W	FOL1	1	0.999	0.100	0.964
6	YDL055C	PSA1	1	0.998	0.100	0.952
7	YDR454C	GUK1	1	0.998	0.100	0.941
8	YOR176W	HEM15	1	0.998	0.100	0.962
9	YPL172C	COX10	0	0.997	0.100	0.963
10	YNR043W	MVD1	1	0.997	0.100	0.964
11	YPR183W	DPM1	1	0.997	0.100	0.928
12	YDL103C	QRI1	1	0.996	0.100	0.956
13	YDR047W	HEM12	1	0.996	0.100	0.965
14	YOR278W	HEM4	1	0.996	0.100	0.965
15	YJL167W	ERG20	1	0.995	0.100	0.964
16	YOR168W	GLN4	1	0.995	0.100	0.942
17	YGR267C	FOL2	1	0.995	0.100	0.946
18	YDL205C	HEM3	1	0.995	0.100	0.965
19	YEL058W	PCM1	1	0.995	0.100	0.961
20	YPL117C	IDI1	1	0.995	0.100	0.965
21	YPR033C	HTS1	1	0.995	0.100	0.947
22	YKL024C	URA6	1	0.994	0.100	0.909
23	YFL017C	GNA1	1	0.994	0.100	0.945
24	YNL247W	YNL247W	1	0.994	0.100	0.956
25	YGL040C	HEM2	1	0.994	0.100	0.965
26	YMR208W	ERG12	1	0.994	0.100	0.894
27	YGR185C	TYS1	1	0.994	0.100	0.956
28	YOR074C	CDC21	1	0.994	0.100	0.956
29	YDL045C	FAD1	1	0.993	0.100	0.952
30	YDR044W	HEM13	1	0.993	0.100	0.966
31	YPL160W	CDC60	1	0.992	0.100	0.959
32	YGR264C	MES1	1	0.991	0.100	0.952

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
33	YDR196C	CAB5	1	0.991	0.100	0.956
34	YGR094W	VAS1	1	0.990	0.100	0.951
35	YBL076C	ILS1	1	0.990	0.100	0.959
36	YMR220W	ERG8	1	0.990	0.100	0.965
37	YDR037W	KRS1	1	0.990	0.100	0.960
38	YDR341C	YDR341C	1	0.989	0.100	0.960
39	YLL018C	DPS1	1	0.989	0.100	0.956
40	YOL097C	WRS1	1	0.989	0.100	0.960
41	YDR531W	CAB1	1	0.988	0.100	0.965
42	YHR019C	DED81	1	0.988	0.100	0.960
43	YER014W	HEM14	0	0.986	0.100	0.966
44	YER003C	PMI40	1	0.986	0.100	0.953
45	YHR020W	YHR020W	1	0.986	0.100	0.965
46	YDR373W	FRQ1	1	0.985	-0.182	0.524
47	YIL078W	THS1	1	0.984	0.100	0.962
48	YOR095C	RKI1	1	0.983	0.100	0.948
49	YNL267W	PIK1	1	0.983	-0.182	0.484
50	YOR335C	ALA1	1	0.982	0.100	0.959
51	YHR074W	QNS1	1	0.982	0.100	0.871
52	YGL245W	GUS1	1	0.981	0.100	0.909
53	YDR208W	MSS4	1	0.979	-0.182	0.497
54	YFL045C	SEC53	1	0.976	0.100	0.963
55	YOR236W	DFR1	1	0.975	0.100	0.939
56	YMR113W	FOL3	1	0.975	0.100	0.906
57	YLR060W	FRS1	1	0.975	0.100	0.958
58	YFL022C	FRS2	1	0.975	0.100	0.958
59	YGL047W	ALG13	1	0.972	-0.182	0.491
60	YBR070C	ALG14	1	0.972	-0.182	0.491
61	YKR072C	SIS2	0	0.971	0.100	0.959
62	YKL088W	CAB3	1	0.971	0.100	0.959
63	YOR054C	VHS3	0	0.971	0.100	0.959
64	YER043C	SAH1	1	0.969	0.100	0.949
65	YGL055W	OLE1	1	0.967	0.100	0.957
66	YPL096C-A	ERI1	0	0.963	-0.182	0.420
67	YPL175W	SPT14	1	0.963	-0.182	0.420
68	YPL076W	GPI2	1	0.963	-0.182	0.420
69	YDR437W	GPI19	1	0.963	-0.182	0.420
70	YNL038W	GPI15	1	0.963	-0.182	0.420
71	YGR216C	GPI1	1	0.963	-0.182	0.420
72	YPL252C	YAH1	1	0.960	-0.182	0.250

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
73	YDR376W	ARH1	1	0.960	-0.182	0.250
74	YNR016C	ACC1	1	0.956	0.100	0.936
75	YML126C	ERG13	1	0.954	0.100	0.950
76	YIL048W	NEO1	1	0.951	-0.182	0.296
77	YPL028W	ERG10	1	0.950	0.100	0.918
78	YER026C	CHO1	0	0.938	0.100	0.961
79	YJL101C	GSH1	1	0.934	0.100	0.884
80	YPR165W	RHO1	1	0.932	0.100	0.792
81	YDL141W	BPL1	1	0.931	0.100	0.886
82	YDR232W	HEM1	1	0.930	0.100	0.958
83	YBR029C	CDS1	1	0.927	0.100	0.949
84	YMR101C	SRT1	0	0.919	-0.182	0.301
85	YBR002C	RER2	1	0.919	-0.182	0.301
86	YBR243C	ALG7	1	0.919	-0.182	0.435
87	YJR013W	GPI14	1	0.916	-0.182	0.441
88	YCL052C	PBN1	1	0.916	-0.182	0.441
89	YOL049W	GSH2	1	0.916	0.100	0.933
90	YPL268W	PLC1	0	0.915	-0.182	0.507
91	YDR062W	LCB2	1	0.914	-0.182	0.429
92	YMR296C	LCB1	1	0.914	-0.182	0.429
93	YGR038W	ORM1	0	0.914	-0.182	0.429
94	YBR058C-A	TSC3	0	0.914	-0.182	0.429
95	YLR350W	ORM2	0	0.914	-0.182	0.429
96	YPL227C	ALG5	0	0.909	-0.182	0.440
97	YGL225W	VRG4	1	0.906	-0.182	0.412
98	YMR217W	GUA1	0	0.901	0.100	0.882
99	YPR113W	PIS1	1	0.901	0.100	0.958
100	YBR192W	RIM2	1	0.895	-0.182	0.225
101	YDL166C	FAP7	1	0.890	-0.182	0.514
102	YEL004W	YEA4	0	0.882	-0.182	0.470
103	YOL130W	ALR1	1	0.874	0.100	0.956
104	YLR305C	STT4	1	0.872	-0.182	0.342
105	YKL104C	GFA1	1	0.870	0.100	0.864
106	YJL091C	GWT1	1	0.870	-0.182	0.514
107	YER023W	PRO3	1	0.867	-0.179	0.569
108	YPL069C	BTS1	0	0.863	-0.182	0.534
109	YGR060W	ERG25	1	0.856	-0.182	0.493
110	YKL152C	GPM1	1	0.848	-0.182	0.474
111	YHR144C	DCD1	0	0.839	-0.182	0.542
112	YKL019W	RAM2	1	0.831	-0.182	0.536

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
113	YDL090C	RAM1	0	0.831	-0.182	0.536
114	YBR110W	ALG1	1	0.831	-0.182	0.469
115	YOR143C	THI80	1	0.830	-0.182	0.557
116	YGR277C	CAB4	1	0.826	-0.182	0.582
117	YHR068W	DYS1	1	0.807	-0.182	0.332
118	YCR012W	PGK1	1	0.797	-0.174	0.546
119	YGL009C	LEU1	0	0.789	-0.038	0.860
120	YPL214C	THI6	0	0.789	-0.182	0.536
121	YFL058W	THI5	0	0.778	-0.182	0.534
122	YLR153C	ACS2	1	0.776	-0.182	0.335
123	YKR069W	MET1	0	0.774	-0.182	0.464
124	YNL275W	BOR1	0	0.773	0.100	0.959
125	YGL142C	GPI10	1	0.773	-0.182	0.489
126	YHR042W	NCP1	1	0.769	-0.182	0.466
127	YHR007C	ERG11	1	0.769	-0.182	0.466
128	YPL231W	FAS2	1	0.764	-0.182	0.418
129	YKL182W	FAS1	1	0.764	-0.182	0.418
130	YJR016C	ILV3	1	0.764	-0.048	0.850
131	YDL120W	YFH1	1	0.760	-0.182	0.544
132	YHR190W	ERG9	1	0.758	-0.182	0.544
133	YER141W	COX15	0	0.756	-0.182	0.442
134	YER061C	CEM1	0	0.755	-0.182	0.523
135	YBR265W	TSC10	1	0.752	-0.182	0.484
136	YLR100W	ERG27	1	0.750	-0.182	0.535
137	YBR004C	GPI18	1	0.748	-0.182	0.464
138	YDL210W	UGA4	0	0.747	-0.182	0.483
139	YKL165C	MCD4	1	0.742	-0.182	0.534
140	YIR027C	DAL1	0	0.742	-0.182	0.544
141	YLR056W	ERG3	0	0.738	-0.182	0.539
142	YDL198C	GGC1	0	0.733	-0.182	0.406
143	YJL005W	CYR1	1	0.728	-0.182	0.480
144	YLR133W	CKI1	0	0.720	-0.182	0.517
145	YLR011W	LOT6	0	0.714	-0.182	0.530
146	YDR503C	LPP1	0	0.713	-0.182	0.566
147	YBR111C	YSA1	0	0.711	-0.182	0.549
148	YOR149C	SMP3	1	0.707	-0.182	0.469
149	YNR030W	ALG12	0	0.707	-0.182	0.469
150	YMR298W	LIP1	1	0.705	-0.182	0.462
151	YHL003C	LAG1	0	0.705	-0.182	0.462
152	YKL008C	LAC1	0	0.705	-0.182	0.462

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
153	YNL219C	ALG9	0	0.704	-0.182	0.470
154	YGR170W	PSD2	0	0.703	-0.182	0.526
155	YNL045W	LAP2	0	0.702	-0.182	0.521
156	YMR013C	SEC59	1	0.699	-0.182	0.479
157	YKL060C	FBA1	1	0.696	-0.182	0.428
158	YGR036C	CAX4	0	0.695	-0.182	0.479
159	YOR002W	ALG6	0	0.695	-0.182	0.479
160	YOR067C	ALG8	0	0.695	-0.182	0.479
161	YGR227W	DIE2	0	0.695	-0.182	0.479
162	YCL018W	LEU2	0	0.690	-0.038	0.867
163	YGL001C	ERG26	1	0.686	-0.182	0.550
164	YBR229C	ROT2	0	0.678	-0.182	0.545
165	YKL035W	UGP1	1	0.677	-0.182	0.314
166	YJR139C	HOM6	0	0.668	-0.113	0.729
167	YBR153W	RIB7	1	0.667	-0.182	0.549
168	YER019W	ISC1	0	0.666	-0.182	0.544
169	YGL038C	OCH1	0	0.663	-0.182	0.539
170	YBR256C	RIB5	1	0.657	-0.182	0.539
171	YER052C	HOM3	0	0.651	-0.113	0.697
172	YMR015C	ERG5	0	0.649	-0.182	0.554
173	YDR236C	FMN1	1	0.649	-0.182	0.476
174	YOR283W	YOR283W	0	0.639	-0.182	0.374
175	YNL280C	ERG24	0	0.636	-0.182	0.544
176	YOR347C	PYK2	0	0.636	-0.182	0.247
177	YAL038W	CDC19	1	0.636	-0.182	0.247
178	YGR286C	BIO2	0	0.634	-0.182	0.544
179	YJR057W	CDC8	1	0.634	-0.182	0.534
180	YOL122C	SMF1	0	0.630	0.100	0.958
181	YLR355C	ILV5	1	0.629	-0.048	0.852
182	YPL110C	GDE1	0	0.628	-0.182	0.514
183	YBR003W	COQ1	0	0.627	-0.182	0.554
184	YNR041C	COQ2	0	0.627	-0.182	0.554
185	YMR108W	ILV2	1	0.626	-0.048	0.658
186	YLR142W	PUT1	0	0.626	-0.182	0.549
187	YIL099W	SGA1	0	0.624	-0.182	0.544
188	YOR221C	MCT1	0	0.623	-0.182	0.526
189	YHR063C	PAN5	0	0.623	-0.182	0.554
190	YPL057C	SUR1	0	0.622	-0.182	0.278
191	YBR161W	CSH1	0	0.622	-0.182	0.278
192	YMR301C	ATM1	1	0.617	-0.182	0.515

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
193	YLL061W	MMP1	0	0.617	-0.182	0.515
194	YIL134W	FLX1	0	0.617	-0.182	0.515
195	YLL048C	YBT1	0	0.617	-0.182	0.515
196	YCR075C	ERS1	0	0.617	-0.182	0.515
197	YCR011C	ADP1	0	0.617	-0.182	0.515
198	YBR295W	PCA1	0	0.617	-0.182	0.515
199	YMR056C	AAC1	0	0.612	-0.182	0.102
200	YBR085W	AAC3	0	0.612	-0.182	0.102
201	YBL030C	PET9	1	0.612	-0.182	0.102
202	YIL145C	PAN6	0	0.612	-0.182	0.534
203	YGL154C	LYS5	0	0.612	-0.052	0.832
204	YBR115C	LYS2	0	0.612	-0.052	0.832
205	YMR281W	GPI12	1	0.611	-0.182	0.554
206	YGR202C	PCT1	0	0.608	-0.182	0.504
207	YGR154C	GTO1	0	0.607	-0.182	0.559
208	YBR046C	ZTA1	0	0.607	-0.182	0.559
209	YPL097W	MSY1	0	0.607	-0.182	0.559
210	YPR047W	MSF1	0	0.607	-0.182	0.559
211	YGL067W	NPY1	0	0.607	-0.182	0.559
212	YDL236W	PHO13	0	0.607	-0.182	0.559
213	YJR153W	PGU1	0	0.607	-0.182	0.559
214	YBR222C	PCS60	0	0.607	-0.182	0.559
215	YDR268W	MSW1	0	0.607	-0.182	0.559
216	YGL012W	ERG4	0	0.607	-0.182	0.559
217	YGL156W	AMS1	0	0.607	-0.182	0.559
218	YBR011C	IPP1	1	0.607	-0.182	0.559
219	YLR151C	PCD1	0	0.607	-0.182	0.559
220	YOL066C	RIB2	1	0.603	-0.182	0.554
221	YBR126C	TPS1	0	0.603	0.100	0.850
222	YDR074W	TPS2	0	0.603	0.100	0.850
223	YEL017C-A	PMP2	0	0.603	-0.182	0.239
224	YCR024C-A	PMP1	0	0.603	-0.182	0.239
225	YGR191W	HIP1	1	0.602	-0.182	0.341
226	YDR305C	HNT2	0	0.597	-0.182	0.555
227	YIL043C	CBR1	0	0.596	-0.182	0.352
228	YML125C	PGA3	1	0.596	-0.182	0.352
229	YAL026C	DRS2	0	0.595	-0.182	0.346
230	YNL129W	NRK1	0	0.593	-0.182	0.544
231	YIR034C	LYS1	0	0.587	-0.052	0.745
232	YGR180C	RNR4	0	0.585	-0.182	0.040

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
233	YER070W	RNR1	0	0.585	-0.182	0.040
234	YJL026W	RNR2	1	0.585	-0.182	0.040
235	YIL066C	RNR3	0	0.585	-0.182	0.040
236	YGL008C	PMA1	1	0.579	-0.182	0.244
237	YPL036W	PMA2	0	0.579	-0.182	0.244
238	YKR067W	GPT2	0	0.579	0.100	0.934
239	YER087W	AIM10	0	0.579	-0.182	0.554
240	YNR057C	BIO4	0	0.578	-0.182	0.549
241	YPR118W	MRI1	0	0.578	-0.182	0.549
242	YGL065C	ALG2	1	0.577	-0.182	0.550
243	YNL048W	ALG11	0	0.577	-0.182	0.550
244	YDR530C	APA2	0	0.574	-0.182	0.554
245	YNR012W	URK1	0	0.574	-0.182	0.387
246	YDR173C	ARG82	0	0.558	-0.182	0.552
247	YBR159W	IFA38	0	0.557	-0.181	0.556
248	YGL027C	CWH41	0	0.555	-0.182	0.554
249	YDR093W	DNF2	0	0.552	-0.182	0.500
250	YIR029W	DAL2	0	0.552	-0.182	0.519
251	YGR007W	ECT1	0	0.550	-0.182	0.494
252	YJL100W	LSB6	0	0.549	-0.182	0.412
253	YJL200C	ACO2	0	0.548	-0.182	0.511
254	YEL021W	URA3	0	0.548	-0.182	0.509
255	YHR072W	ERG7	1	0.548	-0.182	0.539
256	YGR046W	TAM41	1	0.546	-0.182	0.544
257	YKL087C	CYT2	0	0.546	-0.182	0.399
258	YAL039C	CYC3	0	0.546	-0.182	0.399
259	YML120C	NDI1	0	0.544	-0.182	0.338
260	YDR294C	DPL1	0	0.543	-0.182	0.540
261	YDR321W	ASP1	0	0.542	-0.182	0.459
262	YDR050C	TPI1	1	0.542	-0.182	0.499
263	YGR234W	YHB1	0	0.540	-0.182	0.560
264	YGR175C	ERG1	1	0.540	-0.182	0.560
265	YDR036C	EHD3	0	0.540	-0.182	0.560
266	YIR028W	DAL4	0	0.536	-0.182	0.510
267	YKL055C	OAR1	0	0.535	-0.182	0.546
268	YBR026C	ETR1	0	0.535	-0.182	0.546
269	YEL006W	YEA6	0	0.527	-0.182	0.372
270	YJL097W	PHS1	1	0.525	-0.181	0.565
271	YJR152W	DAL5	0	0.525	-0.182	0.422
272	YBR235W	VHC1	0	0.524	-0.182	0.475

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
273	YBR196C	PGI1	1	0.522	-0.182	0.458
274	YKL188C	PXA2	0	0.514	-0.182	0.471
275	YPL147W	PXA1	0	0.514	-0.182	0.471
276	YOL065C	INP54	0	0.509	-0.182	0.562
277	YDR403W	DIT1	0	0.504	-0.182	0.413
278	YLR189C	ATG26	0	0.501	-0.182	0.405
279	YOR316C	COT1	0	0.500	-0.182	0.367
280	YMR243C	ZRC1	0	0.500	-0.182	0.367
281	YDR071C	PAA1	0	0.498	-0.182	0.415
282	YCL030C	HIS4	0	0.496	-0.182	0.491
283	YMR202W	ERG2	0	0.494	-0.182	0.539
284	YBR252W	DUT1	1	0.493	-0.182	0.385
285	YJR024C	MDE1	0	0.492	-0.182	0.544
286	YGL169W	SUA5	1	0.490	-0.182	0.479
287	YBR038W	CHS2	1	0.487	-0.182	0.210
288	YBR023C	CHS3	0	0.487	-0.182	0.210
289	YNL192W	CHS1	0	0.487	-0.182	0.210
290	YDR302W	GPI11	1	0.485	-0.182	0.399
291	YLL031C	GPI13	1	0.485	-0.182	0.399
292	YNL073W	MSK1	0	0.482	-0.182	0.554
293	YHR091C	MSR1	0	0.482	-0.182	0.554
294	YOL143C	RIB4	0	0.482	-0.182	0.549
295	YMR009W	ADI1	0	0.480	-0.182	0.461
296	YEL038W	UTR4	0	0.480	-0.182	0.461
297	YGR157W	CHO2	0	0.476	0.100	0.962
298	YGR240C	PFK1	0	0.476	-0.182	0.481
299	YMR205C	PFK2	0	0.476	-0.182	0.481
300	YDR147W	EKI1	0	0.474	-0.182	0.394
301	YCL038C	ATG22	0	0.468	-0.182	0.423
302	YHR067W	HTD2	0	0.461	-0.182	0.551
303	YIL155C	GUT2	0	0.461	-0.182	0.554
304	YNL141W	AAH1	0	0.460	-0.182	0.506
305	YHR044C	DOG1	0	0.457	-0.182	0.409
306	YHR043C	DOG2	0	0.457	-0.182	0.409
307	YER170W	ADK2	0	0.457	-0.182	0.550
308	YBR022W	POA1	0	0.457	-0.182	0.404
309	YMR087W	YMR087W	0	0.457	-0.182	0.404
310	YOL058W	ARG1	0	0.456	-0.158	0.585
311	YDR017C	KCS1	0	0.456	-0.182	0.552
312	YOR360C	PDE2	0	0.455	-0.182	0.422

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
313	YJR078W	BNA2	0	0.454	-0.182	0.553
314	YIL006W	YIA6	0	0.453	-0.182	0.481
315	YOR128C	ADE2	0	0.452	-0.182	0.544
316	YKL094W	YJU3	0	0.449	-0.182	0.518
317	YGR065C	VHT1	1	0.446	-0.182	0.505
318	YBL033C	RIB1	0	0.446	-0.182	0.435
319	YNL169C	PSD1	0	0.446	-0.182	0.577
320	YDR353W	TRR1	1	0.442	-0.182	0.255
321	YKL085W	MDH1	0	0.440	-0.182	0.520
322	YNR056C	BIO5	0	0.438	-0.182	0.510
323	YCL004W	PGS1	0	0.437	-0.182	0.535
324	YDL015C	TSC13	1	0.436	-0.181	0.506
325	YLR410W	VIP1	0	0.435	-0.182	0.537
326	YOR163W	DDP1	0	0.435	-0.182	0.537
327	YDR234W	LYS4	0	0.433	-0.052	0.840
328	YNL277W	MET2	0	0.433	-0.182	0.425
329	YGR096W	TPC1	0	0.431	-0.182	0.510
330	YKR031C	SPO14	0	0.428	-0.182	0.552
331	YIL094C	LYS12	0	0.423	-0.052	0.844
332	YCR034W	ELO2	0	0.420	-0.181	0.484
333	YLL012W	YEH1	0	0.416	-0.182	0.509
334	YDR402C	DIT2	0	0.415	-0.182	0.428
335	YKL103C	APE1	0	0.413	-0.182	0.530
336	YMR226C	YMR226C	0	0.411	-0.182	0.475
337	YHR128W	FUR1	1	0.410	-0.182	0.405
338	YPL087W	YDC1	0	0.407	-0.182	0.498
339	YBL091C	MAP2	0	0.406	-0.182	0.404
340	YLR244C	MAP1	0	0.406	-0.182	0.404
341	Q0080	ATP8	0	0.404	-0.006	0.447
342	YOL077W-A	ATP19	0	0.404	-0.006	0.447
343	YDR377W	ATP17	0	0.404	-0.006	0.447
344	Q0130	OLI1	0	0.404	-0.006	0.447
345	YBL099W	ATP1	0	0.404	-0.006	0.447
346	YML081C-A	ATP18	0	0.404	-0.006	0.447
347	YBR039W	ATP3	0	0.404	-0.006	0.447
348	YPL271W	ATP15	0	0.404	-0.006	0.447
349	YDL004W	ATP16	1	0.404	-0.006	0.447
350	YKL016C	ATP7	0	0.404	-0.006	0.447
351	YDR298C	ATP5	0	0.404	-0.006	0.447
352	YPL078C	ATP4	0	0.404	-0.006	0.447

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
353	YLR295C	ATP14	0	0.404	-0.006	0.447
354	YJR121W	ATP2	0	0.404	-0.006	0.447
355	YPR020W	ATP20	0	0.404	-0.006	0.447
356	YDR322C-A	TIM11	0	0.404	-0.006	0.447
357	Q0085	ATP6	0	0.404	-0.006	0.447
358	YKL212W	SAC1	0	0.401	-0.182	0.241
359	YFR030W	MET10	0	0.400	-0.182	0.476
360	YJR137C	MET5	0	0.400	-0.182	0.476
361	YHR018C	ARG4	0	0.400	-0.158	0.590
362	YLR020C	YEH2	0	0.399	-0.182	0.535
363	YJR131W	MNS1	0	0.399	-0.182	0.393
364	YLR057W	MNL2	0	0.399	-0.182	0.393
365	YGL037C	PNC1	0	0.392	-0.182	0.534
366	YDR058C	TGL2	0	0.391	-0.182	0.562
367	YIL010W	DOT5	0	0.391	-0.182	0.362
368	YDR297W	SUR2	0	0.391	-0.182	0.496
369	YPR081C	GRS2	0	0.389	-0.182	0.383
370	YBR121C	GRS1	0	0.389	-0.182	0.383
371	YGR192C	TDH3	0	0.386	-0.182	0.226
372	YJL052W	TDH1	0	0.386	-0.182	0.226
373	YJR009C	TDH2	0	0.386	-0.182	0.226
374	YPR160W	GPH1	0	0.385	-0.182	0.509
375	YOR108W	LEU9	0	0.382	-0.182	0.320
376	YDL219W	DTD1	0	0.381	-0.182	0.475
377	YDR315C	IPK1	0	0.379	-0.182	0.554
378	YNL241C	ZWF1	0	0.378	-0.182	0.518
379	YBR018C	GAL7	0	0.374	-0.182	0.402
380	YER055C	HIS1	0	0.373	-0.182	0.474
381	YPR026W	ATH1	0	0.370	-0.182	0.399
382	YBR001C	NTH2	0	0.370	-0.182	0.399
383	YMR099C	YMR099C	0	0.369	-0.182	0.554
384	YDR023W	SES1	1	0.369	-0.182	0.378
385	YHR011W	DIA4	0	0.369	-0.182	0.378
386	YDR539W	FDC1	0	0.369	-0.182	0.506
387	YHR025W	THR1	0	0.368	-0.113	0.745
388	YNR050C	LYS9	0	0.362	-0.052	0.658
389	YGR260W	TNA1	0	0.361	-0.182	0.505
390	YML008C	ERG6	0	0.361	-0.182	0.459
391	YLL041C	SDH2	0	0.359	-0.177	0.142
392	YKL141W	SDH3	1	0.359	-0.177	0.142

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
393	YBR042C	CST26	0	0.359	-0.182	0.500
394	YGR208W	SER2	0	0.358	-0.180	0.523
395	YEL029C	BUD16	0	0.357	-0.182	0.544
396	YNR001C	CIT1	0	0.356	-0.182	0.296
397	YLR017W	MEU1	0	0.355	-0.182	0.509
398	YBR036C	CSG2	0	0.351	-0.182	0.228
399	YLR245C	CDD1	0	0.349	-0.182	0.515
400	YKL067W	YNK1	0	0.349	0.100	0.806
401	YBR176W	ECM31	0	0.339	-0.182	0.504
402	YBR084W	MIS1	0	0.336	-0.182	0.516
403	YBR183W	YPC1	0	0.335	-0.182	0.499
404	YOL011W	PLB3	0	0.334	-0.182	0.448
405	YNR058W	BIO3	0	0.333	-0.182	0.474
406	YCL047C	POF1	0	0.332	-0.182	0.404
407	YDR487C	RIB3	1	0.327	-0.182	0.464
408	YJR019C	TES1	0	0.326	-0.182	0.540
409	YNL202W	SPS19	0	0.326	-0.182	0.550
410	YOR180C	DCI1	0	0.326	-0.182	0.550
411	YPR062W	FCY1	0	0.324	-0.182	0.505
412	YGR194C	XKS1	0	0.322	-0.182	0.539
413	YLR377C	FBP1	0	0.322	-0.182	0.474
414	YFR044C	DUG1	0	0.322	-0.182	0.401
415	YJL196C	ELO1	0	0.320	-0.182	0.530
416	YGL125W	MET13	0	0.316	-0.173	0.425
417	YNL036W	NCE103	1	0.315	-0.182	0.555
418	YEL046C	GLY1	0	0.315	-0.182	0.397
419	YDR009W	GAL3	0	0.313	-0.182	0.549
420	YML035C	AMD1	0	0.313	-0.182	0.509
421	YAR015W	ADE1	0	0.313	-0.182	0.479
422	YBL013W	FMT1	0	0.311	-0.182	0.514
423	YDR178W	SDH4	0	0.308	-0.182	0.139
424	YJL045W	YJL045W	0	0.308	-0.182	0.139
425	YOR155C	ISN1	0	0.304	-0.182	0.499
426	YLR109W	AHP1	0	0.302	-0.182	0.371
427	YPL111W	CAR1	0	0.301	-0.182	0.464
428	YLR372W	ELO3	0	0.300	-0.182	0.474
429	YKL184W	SPE1	0	0.299	-0.182	0.514
430	YNL220W	ADE12	0	0.295	-0.182	0.340
431	YOR209C	NPT1	0	0.294	-0.182	0.417
432	YMR162C	DNF3	0	0.294	-0.182	0.331

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
433	YBR147W	RTC2	0	0.294	-0.182	0.248
434	YCR036W	RBK1	0	0.292	-0.182	0.470
435	YIL023C	YKE4	0	0.288	-0.182	0.505
436	YKR093W	PTR2	0	0.286	-0.182	0.303
437	YJR133W	XPT1	0	0.286	-0.182	0.454
438	YIL124W	AYR1	0	0.283	-0.182	0.488
439	YIL160C	POT1	0	0.282	-0.181	0.549
440	YLR284C	ECI1	0	0.282	-0.182	0.540
441	YNR027W	BUD17	0	0.282	-0.182	0.552
442	YBR041W	FAT1	0	0.281	-0.161	0.205
443	YGR171C	MSM1	0	0.280	-0.182	0.549
444	YDR400W	URH1	0	0.277	-0.182	0.455
445	YKL127W	PGM1	0	0.276	-0.182	0.295
446	YMR105C	PGM2	0	0.276	-0.182	0.295
447	YER042W	MXR1	0	0.276	-0.182	0.415
448	YKL069W	YKL069W	0	0.276	-0.182	0.415
449	YBR263W	SHM1	0	0.275	-0.182	0.504
450	YGL234W	ADE5,7	0	0.275	-0.182	0.460
451	YCL009C	ILV6	0	0.274	-0.182	0.332
452	YLR164W	SHH4	0	0.274	-0.182	0.142
453	YKL148C	SDH1	0	0.274	-0.182	0.142
454	YOR388C	FDH1	0	0.274	-0.182	0.548
455	YGL026C	TRP5	0	0.273	-0.182	0.363
456	YBR034C	HMT1	0	0.273	-0.182	0.425
457	YKL001C	MET14	0	0.271	-0.182	0.534
458	YPL023C	MET12	0	0.269	-0.182	0.410
459	YJL153C	INO1	0	0.269	-0.182	0.484
460	YOR175C	ALE1	0	0.269	-0.182	0.494
461	YDL100C	GET3	0	0.269	-0.182	0.597
462	YPL206C	PGC1	0	0.268	-0.182	0.550
463	YHR100C	GEP4	0	0.268	-0.182	0.550
464	YIL002C	INP51	0	0.268	-0.182	0.169
465	YJL121C	RPE1	0	0.265	-0.182	0.546
466	YBL064C	PRX1	0	0.265	-0.182	0.501
467	YDL238C	GUD1	0	0.264	-0.182	0.529
468	YDR256C	CTA1	0	0.263	-0.182	0.583
469	YKR043C	SHB17	0	0.262	-0.182	0.534
470	YCR083W	TRX3	0	0.261	-0.182	0.488
471	YJR110W	YMR1	0	0.260	-0.182	0.245
472	YDR158W	HOM2	0	0.259	-0.113	0.749

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
473	YJL088W	ARG3	0	0.259	-0.158	0.619
474	YMR250W	GAD1	0	0.259	-0.182	0.293
475	YBR180W	DTR1	0	0.259	-0.182	0.505
476	YLR286C	CTS1	0	0.259	-0.182	0.385
477	YMR267W	PPA2	0	0.258	-0.179	0.578
478	YGL205W	POX1	0	0.258	-0.181	0.544
479	YHR046C	INM1	0	0.256	-0.182	0.394
480	YDR287W	INM2	0	0.256	-0.182	0.394
481	YDR226W	ADK1	0	0.250	-0.182	0.435
482	YDR354W	TRP4	0	0.250	-0.182	0.459
483	YJR001W	AVT1	0	0.249	-0.182	0.299
484	YOR136W	IDH2	0	0.247	-0.180	0.451
485	YNL037C	IDH1	0	0.247	-0.180	0.451
486	YJR010W	MET3	0	0.246	-0.182	0.529
487	YAL023C	PMT2	0	0.244	-0.182	0.151
488	YDL095W	PMT1	0	0.244	-0.182	0.151
489	YMR272C	SCS7	0	0.243	-0.182	0.525
490	YKL194C	MST1	0	0.241	-0.182	0.549
491	YLR351C	NIT3	0	0.240	-0.182	0.435
492	YBL042C	FUI1	0	0.235	-0.182	0.486
493	YKR097W	PCK1	0	0.232	-0.182	0.469
494	YAL022C	FUN26	0	0.230	-0.182	0.466
495	YMR165C	PAH1	0	0.226	-0.182	0.518
496	YGR256W	GND2	0	0.226	-0.182	0.398
497	YHR183W	GND1	0	0.226	-0.182	0.398
498	YCR024C	SLM5	0	0.226	-0.182	0.554
499	YOR306C	MCH5	0	0.225	-0.182	0.524
500	YKL215C	OXP1	0	0.225	-0.182	0.301
501	YLR231C	BNA5	0	0.223	-0.182	0.502
502	YBL011W	SCT1	0	0.223	-0.182	0.359
503	YEL041W	YEF1	0	0.222	-0.182	0.363
504	YJR049C	UTR1	0	0.222	-0.182	0.363
505	YGR155W	CYS4	0	0.221	-0.182	0.425
506	YMR006C	PLB2	0	0.217	-0.182	0.259
507	YIL013C	PDR11	0	0.215	-0.182	0.112
508	YOR011W	AUS1	0	0.215	-0.182	0.112
509	YDR127W	ARO1	0	0.214	-0.182	0.504
510	YOL052C	SPE2	0	0.213	-0.182	0.474
511	YNL268W	LYP1	0	0.212	-0.182	0.370
512	YJR105W	ADO1	0	0.212	-0.182	0.582

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
513	YPR127W	YPR127W	0	0.210	-0.182	0.583
514	YLR354C	TAL1	0	0.207	-0.182	0.309
515	YGR043C	NQM1	0	0.207	-0.182	0.309
516	YPR139C	LOA1	0	0.206	-0.182	0.363
517	YKR080W	MTD1	0	0.203	-0.182	0.572
518	YPL059W	GRX5	0	0.202	-0.182	0.502
519	YER183C	FAU1	0	0.201	-0.182	0.462
520	YDR456W	NHX1	0	0.201	-0.182	0.379
521	YPL274W	SAM3	0	0.200	-0.182	0.385
522	YLR382C	NAM2	0	0.199	-0.182	0.554
523	YPL040C	ISM1	0	0.199	-0.182	0.554
524	YDL022W	GPD1	0	0.198	-0.182	0.413
525	YOR100C	CRC1	0	0.197	-0.182	0.497
526	YGL077C	HNM1	0	0.197	-0.182	0.323
527	YNL117W	MLS1	0	0.197	-0.182	0.322
528	YAL054C	ACS1	0	0.196	-0.182	0.272
529	YML075C	HMG1	0	0.194	-0.182	0.437
530	YLR450W	HMG2	0	0.194	-0.182	0.437
531	YLR237W	THI7	0	0.194	-0.182	0.268
532	YOR071C	NRT1	0	0.194	-0.182	0.268
533	YOR192C	THI72	0	0.194	-0.182	0.268
534	YPL244C	HUT1	0	0.192	-0.182	0.505
535	YPR035W	GLN1	1	0.191	-0.182	0.285
536	YDR408C	ADE8	0	0.190	-0.182	0.499
537	YLR172C	DPH5	0	0.189	-0.182	0.479
538	YHR109W	CTM1	0	0.189	-0.182	0.479
539	YLR285W	NNT1	0	0.187	-0.182	0.464
540	YNL009W	IDP3	0	0.183	-0.182	0.554
541	YLR420W	URA4	0	0.182	-0.182	0.549
542	YIL020C	HIS6	0	0.182	-0.182	0.549
543	YFR025C	HIS2	0	0.181	-0.182	0.549
544	YOR040W	GLO4	0	0.181	-0.182	0.529
545	YKR009C	FOX2	0	0.179	-0.181	0.603
546	YCR098C	GIT1	0	0.179	-0.182	0.339
547	YGR248W	SOL4	0	0.179	-0.182	0.422
548	YHR163W	SOL3	0	0.179	-0.182	0.422
549	YHR106W	TRR2	0	0.176	-0.182	0.382
550	YMR278W	PRM15	0	0.176	-0.182	0.259
551	YCR028C	FEN2	0	0.176	-0.182	0.549
552	YKL140W	TGL1	0	0.174	-0.182	0.365

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
553	YOL064C	MET22	0	0.174	-0.182	0.582
554	YIL009W	FAA3	0	0.173	-0.182	0.334
555	YAL012W	CYS3	0	0.172	-0.182	0.394
556	YOR109W	INP53	0	0.169	-0.182	0.088
557	YNL106C	INP52	0	0.169	-0.182	0.088
558	YER081W	SER3	0	0.169	-0.182	0.299
559	YIL074C	SER33	0	0.169	-0.182	0.299
560	YGL248W	PDE1	0	0.168	-0.182	0.404
561	YJR040W	GEF1	0	0.164	-0.182	0.505
562	YGR144W	THI4	0	0.164	-0.182	0.426
563	YOR253W	NAT5	0	0.164	-0.182	0.339
564	YHR013C	ARD1	0	0.164	-0.182	0.339
565	YDL040C	NAT1	0	0.164	-0.182	0.339
566	YMR289W	ABZ2	0	0.163	-0.182	0.410
567	YLR070C	XYL2	0	0.162	-0.182	0.539
568	YEL031W	SPF1	0	0.162	-0.182	0.510
569	YOR311C	DGK1	0	0.162	-0.182	0.512
570	YML056C	IMD4	0	0.162	-0.182	0.276
571	YHR216W	IMD2	0	0.162	-0.182	0.276
572	YLR432W	IMD3	0	0.162	-0.182	0.276
573	YDR007W	TRP1	0	0.162	-0.182	0.549
574	YLR359W	ADE13	1	0.162	-0.182	0.510
575	YGR204W	ADE3	0	0.161	-0.182	0.465
576	YGL196W	DSD1	0	0.159	-0.182	0.415
577	YDR399W	HPT1	0	0.159	-0.182	0.383
578	YDR428C	BNA7	0	0.156	-0.182	0.528
579	YCR068W	ATG15	0	0.154	-0.182	0.360
580	YLR138W	NHA1	0	0.154	-0.182	0.390
581	YBR006W	UGA2	0	0.151	-0.182	0.499
582	YLR180W	SAM1	0	0.151	-0.182	0.311
583	YDR502C	SAM2	0	0.151	-0.182	0.311
584	YPL148C	PPT2	0	0.150	-0.182	0.597
585	YNL316C	PHA2	0	0.148	-0.182	0.534
586	YBR132C	AGP2	0	0.147	-0.182	0.444
587	YIR032C	DAL3	0	0.145	-0.182	0.500
588	YJR051W	OSM1	0	0.143	-0.182	0.489
589	YKL174C	TPO5	0	0.143	-0.182	0.461
590	YJL134W	LCB3	0	0.143	-0.182	0.351
591	YLR260W	LCB5	0	0.143	-0.182	0.351
592	YKR053C	YSR3	0	0.143	-0.182	0.351

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
593	YOR171C	LCB4	0	0.143	-0.182	0.351
594	YDR497C	ITR1	0	0.141	-0.182	0.399
595	YOL103W	ITR2	0	0.141	-0.182	0.399
596	YBR166C	TYR1	0	0.141	-0.182	0.524
597	YMR008C	PLB1	0	0.137	-0.182	0.463
598	YNL270C	ALP1	0	0.137	-0.182	0.252
599	YEL063C	CAN1	0	0.137	-0.182	0.252
600	YLR303W	MET17	0	0.136	-0.182	0.307
601	YFR049W	YMR31	0	0.134	-0.182	0.227
602	YLR174W	IDP2	0	0.134	-0.182	0.463
603	YGR061C	ADE6	0	0.133	-0.182	0.277
604	YKL029C	MAE1	0	0.133	-0.179	0.469
605	YER091C	MET6	0	0.133	-0.173	0.421
606	YOR334W	MRS2	0	0.132	-0.182	0.358
607	YPL060W	MFM1	0	0.132	-0.182	0.358
608	YHL016C	DUR3	0	0.132	-0.182	0.429
609	YOL092W	YPQ1	0	0.131	-0.182	0.277
610	YDR352W	YPQ2	0	0.131	-0.182	0.277
611	YGL224C	SDT1	0	0.128	-0.182	0.342
612	YHL032C	GUT1	0	0.127	-0.182	0.489
613	YDL024C	DIA3	0	0.127	-0.182	0.393
614	YOR202W	HIS3	0	0.127	-0.182	0.549
615	YER069W	ARG5,6	0	0.126	-0.158	0.648
616	YER086W	ILV1	0	0.126	-0.182	0.529
617	YER015W	FAA2	0	0.125	-0.182	0.353
618	YBR208C	DUR1,2	0	0.125	-0.182	0.485
619	YML042W	CAT2	0	0.124	-0.182	0.393
620	YKL216W	URA1	0	0.124	-0.182	0.474
621	YHR008C	SOD2	0	0.124	-0.182	0.529
622	YGR012W	YGR012W	0	0.124	-0.182	0.534
623	YGL148W	ARO2	0	0.123	-0.182	0.534
624	YNL274C	GOR1	0	0.121	-0.182	0.529
625	YMR210W	YMR210W	0	0.121	-0.182	0.489
626	YBR291C	CTP1	0	0.121	-0.182	0.366
627	YDL142C	CRD1	0	0.120	-0.182	0.560
628	YOR161C	PNS1	0	0.119	-0.182	0.326
629	YDR270W	CCC2	0	0.118	-0.182	0.510
630	YKL218C	SRY1	0	0.118	-0.182	0.524
631	YOR323C	PRO2	0	0.118	-0.182	0.592
632	YJR143C	PMT4	0	0.117	-0.182	0.161

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
633	YDL093W	PMT5	0	0.117	-0.182	0.161
634	YOR321W	PMT3	0	0.117	-0.182	0.161
635	YLR209C	PNP1	0	0.115	-0.182	0.306
636	YCR053W	THR4	0	0.114	-0.113	0.596
637	YCR005C	CIT2	0	0.114	-0.182	0.577
638	YKL192C	ACP1	0	0.114	-0.182	0.422
639	YMR246W	FAA4	0	0.113	-0.182	0.104
640	YPL104W	MSD1	0	0.110	-0.182	0.549
641	YMR207C	HFA1	0	0.110	-0.182	0.449
642	YKL150W	MCR1	0	0.109	-0.182	0.280
643	YOL126C	MDH2	0	0.108	-0.182	0.489
644	YBR213W	MET8	0	0.108	-0.182	0.495
645	YJL129C	TRK1	0	0.105	-0.182	0.404
646	YPL092W	SSU1	0	0.103	-0.182	0.485
647	YMR145C	NDE1	0	0.101	-0.182	0.256
648	YDL085W	NDE2	0	0.101	-0.182	0.256
649	YGL006W	PMC1	0	0.099	-0.182	0.505
650	YGL167C	PMR1	0	0.099	-0.182	0.505
651	YFR015C	GSY1	0	0.099	-0.182	0.119
652	YLR258W	GSY2	0	0.099	-0.182	0.119
653	YKR058W	GLG1	0	0.099	-0.182	0.119
654	YJL137C	GLG2	0	0.099	-0.182	0.119
655	YBR149W	ARA1	0	0.097	-0.182	0.539
656	YMR041C	ARA2	0	0.097	-0.182	0.539
657	YBR293W	VBA2	0	0.097	-0.182	0.227
658	YMR241W	YHM2	0	0.095	-0.182	0.306
659	YOR348C	PUT4	0	0.095	-0.182	0.264
660	YGL255W	ZRT1	0	0.094	-0.182	0.404
661	YLR130C	ZRT2	0	0.094	-0.182	0.404
662	YJL126W	NIT2	0	0.094	-0.182	0.307
663	YOL151W	GRE2	0	0.093	-0.182	0.498
664	YOR317W	FAA1	0	0.091	-0.182	0.169
665	YJR025C	BNA1	0	0.091	-0.182	0.597
666	YER065C	ICL1	0	0.089	-0.182	0.459
667	YBL080C	PET112	0	0.089	-0.182	0.514
668	YOL033W	MSE1	0	0.089	-0.182	0.514
669	YBL098W	BNA4	0	0.089	-0.182	0.587
670	YPR128C	ANT1	0	0.087	-0.182	0.538
671	YDL066W	IDP1	0	0.086	-0.182	0.497
672	YPR002W	PDH1	0	0.085	-0.182	0.592

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
673	YNL325C	FIG4	0	0.085	-0.182	0.445
674	YLR386W	VAC14	0	0.085	-0.182	0.445
675	YNL054W	VAC7	0	0.085	-0.182	0.445
676	YFR021W	ATG18	0	0.085	-0.182	0.445
677	YNR033W	ABZ1	0	0.083	-0.182	0.273
678	YNL003C	PET8	0	0.081	-0.182	0.411
679	YER119C	AVT6	0	0.081	-0.182	0.247
680	YJR130C	STR2	0	0.079	-0.182	0.349
681	YLR348C	DIC1	0	0.079	-0.182	0.360
682	YDR284C	DPP1	0	0.078	-0.182	0.581
683	YDL078C	MDH3	0	0.075	-0.181	0.590
684	YML059C	NTE1	0	0.074	-0.182	0.542
685	YLR328W	NMA1	0	0.074	-0.182	0.345
686	YPR060C	ARO7	0	0.073	-0.182	0.529
687	YFR047C	BNA6	0	0.071	-0.182	0.448
688	YEL066W	HPA3	0	0.071	-0.182	0.308
689	YPR193C	HPA2	0	0.071	-0.182	0.308
690	YGR143W	SKN1	0	0.071	-0.182	0.282
691	YPR159W	KRE6	0	0.071	-0.182	0.282
692	YCL064C	CHA1	0	0.070	-0.182	0.306
693	YOR120W	GCY1	0	0.066	-0.182	0.494
694	YJR095W	SFC1	0	0.065	-0.182	0.392
695	YKR052C	MRS4	0	0.064	-0.182	0.348
696	YJL133W	MRS3	0	0.064	-0.182	0.348
697	YDR452W	PPN1	0	0.064	-0.182	0.095
698	YJR073C	OPI3	0	0.063	0.100	0.963
699	YMR062C	ARG7	0	0.060	-0.158	0.483
700	YIL107C	PFK26	0	0.060	-0.182	0.339
701	YOL136C	PFK27	0	0.060	-0.182	0.339
702	YJL155C	FBP26	0	0.060	-0.182	0.339
703	YMR261C	TPS3	0	0.060	-0.182	0.237
704	YML100W	TSL1	0	0.060	-0.182	0.237
705	YGL186C	TPN1	0	0.059	-0.175	0.508
706	YJL012C	VTC4	0	0.059	-0.182	0.505
707	YNL065W	AQR1	0	0.059	-0.182	0.218
708	YBR020W	GAL1	0	0.058	-0.182	0.519
709	YDL215C	GDH2	0	0.058	-0.182	0.265
710	YPL258C	THI21	0	0.057	-0.182	0.267
711	YOL055C	THI20	0	0.057	-0.182	0.267
712	YPR121W	THI22	0	0.057	-0.182	0.267

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
713	YGR088W	CTT1	0	0.056	-0.182	0.494
714	YJR104C	SOD1	0	0.056	-0.182	0.494
715	YGR032W	GSC2	0	0.055	-0.182	0.172
716	YMR306W	FKS3	0	0.055	-0.182	0.172
717	YLR342W	FKS1	0	0.055	-0.182	0.172
718	YOL059W	GPD2	0	0.054	-0.182	0.480
719	YML106W	URA5	0	0.053	-0.182	0.330
720	YMR271C	URA10	0	0.053	-0.182	0.330
721	YER024W	YAT2	0	0.052	-0.182	0.300
722	YAR035W	YAT1	0	0.052	-0.182	0.300
723	YBR294W	SUL1	0	0.051	-0.182	0.389
724	YLR092W	SUL2	0	0.051	-0.182	0.389
725	YBL015W	ACH1	0	0.050	-0.182	0.359
726	YAL060W	BDH1	0	0.048	-0.182	0.532
727	YLL057C	JLP1	0	0.048	-0.182	0.354
728	YLL028W	TPO1	0	0.048	-0.182	0.166
729	YOR273C	TPO4	0	0.048	-0.182	0.166
730	YBR035C	PDX3	0	0.047	-0.182	0.458
731	YOL140W	ARG8	0	0.047	-0.158	0.541
732	YPR167C	MET16	0	0.047	-0.182	0.309
733	YHR123W	EPT1	0	0.047	-0.182	0.450
734	YNL130C	CPT1	0	0.046	-0.182	0.320
735	YML086C	ALO1	0	0.044	-0.182	0.484
736	YFL030W	AGX1	0	0.043	-0.182	0.305
737	YBR248C	HIS7	0	0.043	-0.182	0.265
738	YGR010W	NMA2	0	0.042	-0.182	0.367
739	YLL051C	FRE6	0	0.042	-0.182	0.455
740	YPR006C	ICL2	0	0.041	-0.182	0.449
741	YBR021W	FUR4	0	0.040	-0.180	0.528
742	YGR110W	CLD1	0	0.038	-0.182	0.589
743	YGL062W	PYC1	0	0.037	-0.182	0.265
744	YBR218C	PYC2	0	0.037	-0.182	0.265
745	YLR299W	ECM38	0	0.037	-0.182	0.396
746	YNL101W	AVT4	0	0.037	-0.182	0.178
747	YKL146W	AVT3	0	0.037	-0.182	0.178
748	YOR130C	ORT1	0	0.036	-0.158	0.591
749	YKR076W	ECM4	0	0.034	-0.182	0.454
750	YFR019W	FAB1	0	0.034	-0.182	0.455
751	YGR289C	MAL11	0	0.034	-0.182	0.439
752	YGR019W	UGA1	0	0.034	-0.182	0.249

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
753	YMR020W	FMS1	0	0.033	-0.182	0.422
754	YIL125W	KGD1	0	0.033	-0.179	0.124
755	YDR148C	KGD2	0	0.033	-0.179	0.124
756	YPL188W	POS5	0	0.033	-0.182	0.283
757	YDR384C	ATO3	0	0.033	-0.182	0.187
758	YGR121C	MEP1	0	0.033	-0.182	0.187
759	YPR138C	MEP3	0	0.033	-0.182	0.187
760	YNL142W	MEP2	0	0.033	-0.182	0.187
761	YJL068C	YJL068C	0	0.032	-0.182	0.401
762	YDL052C	SLC1	0	0.032	-0.182	0.279
763	YML022W	APT1	0	0.031	-0.182	0.313
764	YDR441C	APT2	0	0.031	-0.182	0.313
765	YER163C	GCG1	0	0.031	-0.182	0.440
766	YJL212C	OPT1	0	0.030	-0.182	0.412
767	YLR089C	ALT1	0	0.029	-0.182	0.372
768	YDL171C	GLT1	0	0.029	-0.182	0.246
769	YLR058C	SHM2	0	0.028	-0.182	0.400
770	YGR177C	ATF2	0	0.028	-0.182	0.102
771	YOR377W	ATF1	0	0.028	-0.182	0.102
772	YDR035W	ARO3	0	0.027	-0.182	0.295
773	YBR249C	ARO4	0	0.027	-0.182	0.295
774	YDR481C	PHO8	0	0.027	-0.182	0.411
775	YOR245C	DGA1	0	0.026	-0.182	0.248
776	YDR135C	YCF1	0	0.025	-0.182	0.405
777	YLR043C	TRX1	0	0.025	-0.182	0.118
778	YIL116W	HIS5	0	0.024	-0.182	0.257
779	YPR058W	YMC1	0	0.024	-0.182	0.454
780	YIL053W	GPP1	0	0.024	-0.182	0.342
781	YER062C	GPP2	0	0.024	-0.182	0.342
782	YLR028C	ADE16	0	0.023	-0.182	0.311
783	YMR120C	ADE17	0	0.023	-0.182	0.311
784	YAL035W	FUN12	0	0.023	-0.182	0.211
785	YLL001W	DNM1	0	0.023	-0.182	0.211
786	YCR010C	ADY2	0	0.023	-0.182	0.455
787	YBR069C	TAT1	0	0.021	-0.182	0.132
788	YPR140W	TAZ1	0	0.021	-0.182	0.750
789	YBR117C	TKL2	0	0.020	-0.182	0.237
790	YPR074C	TKL1	0	0.020	-0.182	0.237
791	YPL262W	FUM1	0	0.020	-0.176	0.432
792	YLR307W	CDA1	0	0.020	-0.182	0.296

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
793	YLR308W	CDA2	0	0.020	-0.182	0.296
794	YDR453C	TSA2	0	0.020	-0.182	0.292
795	YLR304C	ACO1	0	0.020	-0.182	0.489
796	YDR038C	ENA5	0	0.019	-0.182	0.235
797	YDR039C	ENA2	0	0.019	-0.182	0.235
798	YDR040C	ENA1	0	0.019	-0.182	0.235
799	YOR285W	RDL1	0	0.019	-0.182	0.280
800	YOR286W	RDL2	0	0.019	-0.182	0.280
801	YBR092C	PHO3	0	0.019	-0.182	0.395
802	YEL047C	FRD1	0	0.018	-0.182	0.259
803	YBR128C	ATG14	0	0.018	-0.182	0.169
804	YML054C	CYB2	0	0.017	-0.182	0.139
805	YMR313C	TGL3	0	0.017	-0.182	0.305
806	YPL120W	VPS30	0	0.017	-0.182	0.141
807	YLR438W	CAR2	0	0.016	-0.182	0.246
808	YJR069C	HAM1	0	0.016	-0.182	0.361
809	YDR272W	GLO2	0	0.015	-0.182	0.405
810	YLR360W	VPS38	0	0.015	-0.182	0.179
811	YPR001W	CIT3	0	0.015	-0.182	0.295
812	YNR008W	LRO1	0	0.013	-0.182	0.649
813	YGL080W	MPC1	0	0.013	-0.182	0.198
814	YKR089C	TGL4	0	0.013	-0.182	0.289
815	YKL217W	JEN1	0	0.013	-0.182	0.174
816	YEL042W	GDA1	0	0.013	-0.182	0.377
817	YHR162W	MPC2	0	0.012	-0.182	0.202
818	YGR243W	MPC3	0	0.012	-0.182	0.202
819	YKR066C	CCP1	0	0.012	-0.182	0.230
820	YKL132C	RMA1	0	0.012	-0.182	0.173
821	YOR241W	MET7	0	0.012	-0.182	0.173
822	YIR031C	DAL7	0	0.012	-0.182	0.260
823	YER178W	PDA1	0	0.012	-0.176	0.345
824	YNL071W	LAT1	0	0.012	-0.176	0.345
825	YGR193C	PDX1	0	0.012	-0.176	0.345
826	YBR221C	PDB1	0	0.012	-0.176	0.345
827	YDR300C	PRO1	0	0.011	-0.182	0.339
828	YMR058W	FET3	0	0.011	-0.182	0.357
829	YMR319C	FET4	0	0.011	-0.182	0.357
830	YGL184C	STR3	0	0.011	-0.182	0.245
831	YML004C	GLO1	0	0.010	-0.182	0.425
832	YCL050C	APA1	0	0.010	-0.182	0.311

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
833	YBR097W	VPS15	0	0.009	-0.182	0.118
834	YLR240W	VPS34	0	0.009	-0.182	0.118
835	YPL273W	SAM4	0	0.009	-0.182	0.248
836	YLL062C	MHT1	0	0.009	-0.182	0.248
837	YMR300C	ADE4	0	0.009	-0.182	0.221
838	YPL091W	GLR1	0	0.009	-0.182	0.340
839	YJR159W	SOR1	0	0.007	-0.182	0.438
840	YER060W-A	FCY22	0	0.007	-0.182	0.211
841	YER060W	FCY21	0	0.007	-0.182	0.211
842	YER056C	FCY2	0	0.007	-0.182	0.211
843	YPR156C	TPO3	0	0.007	-0.182	0.079
844	YGR138C	TPO2	0	0.007	-0.182	0.079
845	YHR174W	ENO2	0	0.007	-0.182	0.095
846	YGR254W	ENO1	0	0.007	-0.182	0.095
847	YMR323W	ERR3	0	0.007	-0.182	0.095
848	YOR393W	ERR1	0	0.007	-0.182	0.095
849	YPL281C	ERR2	0	0.007	-0.182	0.095
850	YLL043W	FPS1	0	0.007	-0.182	0.308
851	YGR255C	COQ6	0	0.007	-0.182	0.331
852	YDR204W	COQ4	0	0.007	-0.182	0.331
853	YLR201C	COQ9	0	0.007	-0.182	0.331
854	YOL096C	COQ3	0	0.007	-0.182	0.331
855	YOR125C	CAT5	0	0.007	-0.182	0.331
856	YML110C	COQ5	0	0.007	-0.182	0.331
857	YKL120W	OAC1	0	0.006	-0.182	0.415
858	YHR201C	PPX1	0	0.006	-0.182	0.068
859	YOR251C	TUM1	0	0.006	-0.182	0.293
860	YBR019C	GAL10	0	0.005	-0.182	0.336
861	YMR303C	ADH2	0	0.005	-0.182	0.484
862	YOR142W	LSC1	0	0.005	-0.179	0.330
863	YGR244C	LSC2	0	0.005	-0.179	0.330
864	YAL062W	GDH3	0	0.004	-0.182	0.190
865	YOR375C	GDH1	0	0.004	-0.182	0.190
866	YDL131W	LYS21	0	0.004	-0.182	0.224
867	YDL182W	LYS20	0	0.004	-0.182	0.224
868	YDR380W	ARO10	0	0.004	-0.182	0.553
869	YML070W	DAK1	0	0.004	-0.182	0.215
870	YFL053W	DAK2	0	0.004	-0.182	0.215
871	YJL071W	ARG2	0	0.004	-0.182	0.279
872	YPR184W	GDB1	0	0.004	-0.182	0.368

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
873	YGR124W	ASN2	0	0.003	-0.182	0.187
874	YPR145W	ASN1	0	0.003	-0.182	0.187
875	YBR296C	PHO89	0	0.003	-0.182	0.118
876	YJR103W	URA8	0	0.003	-0.182	0.154
877	YBL039C	URA7	0	0.003	-0.182	0.154
878	YLL015W	BPT1	0	0.003	-0.182	0.280
879	YKL181W	PRS1	0	0.003	-0.182	0.076
880	YDL178W	DLD2	0	0.003	-0.182	0.148
881	YDR001C	NTH1	0	0.003	-0.182	0.368
882	YOL061W	PRS5	0	0.003	-0.182	0.078
883	YER099C	PRS2	0	0.003	-0.182	0.078
884	YBL068W	PRS4	0	0.003	-0.182	0.078
885	YBL045C	COR1	0	0.003	0.044	0.358
886	YPR191W	QCR2	0	0.003	0.044	0.358
887	Q0105	COB	0	0.003	0.044	0.358
888	YGR183C	QCR9	0	0.003	0.044	0.358
889	YEL024W	RIP1	0	0.003	0.044	0.358
890	YFR033C	QCR6	0	0.003	0.044	0.358
891	YDR529C	QCR7	0	0.003	0.044	0.358
892	YHR001W-A	QCR10	0	0.003	0.044	0.358
893	YJL166W	QCR8	0	0.003	0.044	0.358
894	YOR065W	CYT1	0	0.003	0.044	0.358
895	YHL011C	PRS3	0	0.003	-0.182	0.080
896	YOR081C	TGL5	0	0.003	-0.182	0.111
897	YER152C	YER152C	0	0.003	-0.182	0.116
898	YER175C	TMT1	0	0.003	-0.182	0.350
899	YDL080C	THI3	0	0.002	-0.182	0.493
900	YJL130C	URA2	0	0.002	-0.182	0.239
901	YDR538W	PAD1	0	0.002	-0.182	0.384
902	YNL104C	LEU4	0	0.002	-0.182	0.321
903	YJR148W	BAT2	0	0.002	-0.038	0.376
904	YOR184W	SER1	0	0.002	-0.180	0.300
905	YPL095C	EEB1	0	0.002	-0.182	0.365
906	YBR177C	EHT1	0	0.002	-0.182	0.365
907	YDR019C	GCV1	0	0.002	-0.182	0.295
908	YAL044C	GCV3	0	0.002	-0.182	0.295
909	YMR189W	GCV2	0	0.002	-0.182	0.295
910	YPR069C	SPE3	0	0.002	-0.182	0.365
911	YIL162W	SUC2	0	0.002	-0.182	0.425
912	YJR109C	CPA2	0	0.002	-0.182	0.173

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
913	YOR303W	CPA1	0	0.002	-0.182	0.173
914	YNR019W	ARE2	0	0.002	-0.182	0.110
915	YPR021C	AGC1	0	0.002	-0.181	0.231
916	YDL067C	COX9	0	0.002	0.045	0.051
917	Q0045	COX1	0	0.002	0.045	0.051
918	Q0250	COX2	0	0.002	0.045	0.051
919	Q0275	COX3	0	0.002	0.045	0.051
920	YER010C	YER010C	0	0.001	-0.182	0.281
921	YNL334C	SNO2	0	0.001	-0.182	0.097
922	YFL060C	SNO3	0	0.001	-0.182	0.097
923	YOL020W	TAT2	0	0.001	-0.182	0.050
924	YLR146C	SPE4	0	0.001	-0.182	0.356
925	YJL198W	PHO90	0	0.001	-0.182	0.154
926	YCR037C	PHO87	0	0.001	-0.182	0.154
927	YML123C	PHO84	0	0.001	-0.182	0.154
928	YBR204C	LDH1	0	0.001	-0.182	0.078
929	YCL069W	VBA3	0	0.001	-0.182	0.126
930	YMR088C	VBA1	0	0.001	-0.182	0.126
931	YBR281C	DUG2	0	0.001	-0.182	0.209
932	YNL191W	DUG3	0	0.001	-0.182	0.209
933	YDR046C	BAP3	0	0.001	-0.182	0.043
934	YBR068C	BAP2	0	0.001	-0.182	0.043
935	YAR071W	PHO11	0	0.001	-0.182	0.177
936	YBR093C	PHO5	0	0.001	-0.182	0.177
937	YHR215W	PHO12	0	0.001	-0.182	0.177
938	YGL084C	GUP1	0	0.001	-0.182	0.214
939	YPL189W	GUP2	0	0.001	-0.182	0.214
940	YHR051W	COX6	0	0.001	-0.182	0.005
941	YGL191W	COX13	0	0.001	-0.182	0.005
942	YGL187C	COX4	0	0.001	-0.182	0.005
943	YMR256C	COX7	0	0.001	-0.182	0.005
944	YLR395C	COX8	0	0.001	-0.182	0.005
945	YLR038C	COX12	0	0.001	-0.182	0.005
946	YOL157C	IMA2	0	0.001	-0.182	0.371
947	YHR037W	PUT2	0	0.001	-0.182	0.388
948	YPL134C	ODC1	0	0.001	-0.182	0.204
949	YOR222W	ODC2	0	0.001	-0.182	0.204
950	YDR231C	COX20	0	0.001	-0.182	0.006
951	YLL018C-A	COX19	0	0.001	-0.182	0.006
952	YPL132W	COX11	0	0.001	-0.182	0.006

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
953	YJL003W	COX16	0	0.001	-0.182	0.006
954	YHR116W	COX23	0	0.001	-0.182	0.006
955	YGR062C	COX18	0	0.001	-0.182	0.006
956	YHR002W	LEU5	0	0.001	-0.182	0.351
957	YNL052W	COX5A	0	0.001	-0.182	0.005
958	YIL111W	COX5B	0	0.001	-0.182	0.005
959	YMR293C	HER2	0	0.001	-0.182	0.409
960	YOR270C	VPH1	0	0.001	-0.182	0.163
961	YMR054W	STV1	0	0.001	-0.182	0.163
962	YCR048W	ARE1	0	0.001	-0.182	0.183
963	YHR092C	HXT4	0	0.000	-0.182	0.224
964	YLR158C	ASP3-3	0	0.000	-0.182	0.166
965	YLR155C	ASP3-1	0	0.000	-0.182	0.166
966	YLR160C	ASP3-4	0	0.000	-0.182	0.166
967	YLR157C	ASP3-2	0	0.000	-0.182	0.166
968	YFL059W	SNZ3	0	0.000	-0.182	0.150
969	YNL333W	SNZ2	0	0.000	-0.182	0.150
970	YER090W	TRP2	0	0.000	-0.182	0.172
971	YER053C	PIC2	0	0.000	-0.182	0.226
972	YJR077C	MIR1	0	0.000	-0.182	0.226
973	YEL071W	DLD3	0	0.000	-0.182	0.080
974	YDR105C	TMS1	0	0.000	-0.182	0.053
975	YKL106W	AAT1	0	0.000	-0.182	0.301
976	YPL212C	PUS1	0	0.000	-0.182	0.138
977	YNL292W	PUS4	0	0.000	-0.182	0.138
978	YFL001W	DEG1	0	0.000	-0.182	0.138
979	YGL063W	PUS2	0	0.000	-0.182	0.138
980	YER005W	YND1	0	0.000	-0.182	0.085
981	YDL174C	DLD1	0	0.000	-0.182	0.047
982	YFL055W	AGP3	0	0.000	-0.182	0.077
983	YGL253W	HXK2	0	0.000	-0.182	0.095
984	YFR053C	HXK1	0	0.000	-0.182	0.095
985	YGR247W	CPD1	0	0.000	-0.182	0.437
986	YKL211C	TRP3	0	0.000	-0.182	0.167
987	YHR208W	BAT1	0	0.000	-0.059	0.230
988	YDL245C	HXT15	0	0.000	-0.182	0.147
989	YBR299W	MAL32	0	0.000	-0.182	0.195
990	YGR292W	MAL12	0	0.000	-0.182	0.195
991	YDR508C	GNP1	0	0.000	-0.182	0.081
992	YDR342C	HXT7	0	0.000	-0.182	0.138

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
993	YDR533C	HSP31	0	0.000	-0.182	0.152
994	YOR391C	HSP33	0	0.000	-0.182	0.152
995	YDR516C	EMI2	0	0.000	-0.182	0.056
996	YCL040W	GLK1	0	0.000	-0.182	0.056
997	YHL036W	MUP3	0	0.000	-0.182	0.029
998	YGR055W	MUP1	0	0.000	-0.182	0.029
999	YOL068C	HST1	0	0.000	-0.182	0.110
1000	YOR025W	HST3	0	0.000	-0.182	0.110
1001	YPL015C	HST2	0	0.000	-0.182	0.110
1002	YDR191W	HST4	0	0.000	-0.182	0.110
1003	YDL042C	SIR2	0	0.000	-0.182	0.110
1004	YEL051W	VMA8	0	0.000	-0.182	0.134
1005	YBR127C	VMA2	0	0.000	-0.182	0.134
1006	YGR106C	VOA1	0	0.000	-0.182	0.134
1007	YEL027W	VMA3	0	0.000	-0.182	0.134
1008	YHR039C-A	VMA10	0	0.000	-0.182	0.134
1009	YLR447C	VMA6	0	0.000	-0.182	0.134
1010	YKL080W	VMA5	0	0.000	-0.182	0.134
1011	YHR026W	VMA16	0	0.000	-0.182	0.134
1012	YPL234C	VMA11	0	0.000	-0.182	0.134
1013	YGR020C	VMA7	0	0.000	-0.182	0.134
1014	YPR036W	VMA13	0	0.000	-0.182	0.134
1015	YCL005W-A	VMA9	0	0.000	-0.182	0.134
1016	YDL185W	VMA1	0	0.000	-0.182	0.134
1017	YOR332W	VMA4	0	0.000	-0.182	0.134
1018	YER174C	GRX4	0	0.000	-0.182	0.125
1019	YDR098C	GRX3	0	0.000	-0.182	0.125
1020	YGR209C	TRX2	0	0.000	-0.182	0.046
1021	YLR027C	AAT2	0	0.000	-0.125	0.133
1022	YER037W	PHM8	0	0.000	-0.182	0.235
1023	YBR298C	MAL31	0	0.000	-0.182	0.140
1024	YER073W	ALD5	0	0.000	-0.182	0.113
1025	YDR536W	STL1	0	0.000	-0.182	0.102
1026	YCL035C	GRX1	0	0.000	-0.182	0.106
1027	YDR513W	GRX2	0	0.000	-0.182	0.106
1028	YJL060W	BNA3	0	0.000	-0.182	0.104
1029	YPL265W	DIP5	0	0.000	-0.182	0.047
1030	YDL247W	MPH2	0	0.000	-0.182	0.073
1031	YJR160C	MPH3	0	0.000	-0.182	0.073
1032	YCR107W	AAD3	0	0.000	-0.182	0.057

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
1033	YNL331C	AAD14	0	0.000	-0.182	0.057
1034	YDL243C	AAD4	0	0.000	-0.182	0.057
1035	YJR155W	AAD10	0	0.000	-0.182	0.057
1036	YOL165C	AAD15	0	0.000	-0.182	0.057
1037	YOR126C	IAH1	0	0.000	-0.182	0.161
1038	YNR013C	PHO91	0	0.000	-0.182	0.220
1039	YNL318C	HXT14	0	0.000	-0.182	0.091
1040	YEL069C	HXT13	0	0.000	-0.182	0.058
1041	YNR072W	HXT17	0	0.000	-0.182	0.058
1042	YJR158W	HXT16	0	0.000	-0.182	0.058
1043	YLR081W	GAL2	0	0.000	-0.182	0.043
1044	YMR169C	ALD3	0	0.000	-0.182	0.162
1045	YHR104W	GRE3	0	0.000	-0.182	0.212
1046	YDR368W	YPR1	0	0.000	-0.182	0.109
1047	YCR105W	ADH7	0	0.000	-0.182	0.131
1048	YMR318C	ADH6	0	0.000	-0.182	0.131
1049	YEL039C	CYC7	0	0.000	-0.182	0.001
1050	YJR048W	CYC1	0	0.000	-0.182	0.001
1051	YKR039W	GAP1	0	0.000	0.100	0.100
1052	YCL025C	AGP1	0	0.000	-0.182	0.023
1053	YJL219W	HXT9	0	0.000	-0.182	0.034
1054	YOL156W	HXT11	0	0.000	-0.182	0.034
1055	YFL011W	HXT10	0	0.000	-0.182	0.034
1056	YGR282C	BGL2	0	0.000	-0.182	0.073
1057	YLR300W	EXG1	0	0.000	-0.182	0.073
1058	YNR067C	DSE4	0	0.000	-0.182	0.073
1059	YOR190W	SPR1	0	0.000	-0.182	0.073
1060	YDR261C	EXG2	0	0.000	-0.182	0.073
1061	YBR244W	GPX2	0	0.000	-0.182	0.041
1062	YKL026C	GPX1	0	0.000	-0.182	0.041
1063	YNL229C	URE2	0	0.000	-0.182	0.041
1064	YIR037W	HYR1	0	0.000	-0.182	0.041
1065	YLR047C	FRE8	0	0.000	-0.182	0.028
1066	YOL152W	FRE7	0	0.000	-0.182	0.028
1067	YOR381W	FRE3	0	0.000	-0.182	0.028
1068	YNR060W	FRE4	0	0.000	-0.182	0.028
1069	YKL220C	FRE2	0	0.000	-0.182	0.028
1070	YLR214W	FRE1	0	0.000	-0.182	0.028
1071	YOR384W	FRE5	0	0.000	-0.182	0.028
1072	YDR483W	KRE2	0	0.000	-0.182	0.020

Table 9: Full per-gene output of the tool: all 1,107 benchmark genes ranked by full-data GCN essentiality score, with the FBA rule and logistic-regression scores for comparison.

rank	gene	name	essential	GCN	FBA	LR
1073	YPL053C	KTR6	0	0.000	-0.182	0.020
1074	YNL029C	KTR5	0	0.000	-0.182	0.020
1075	YBR199W	KTR4	0	0.000	-0.182	0.020
1076	YJL139C	YUR1	0	0.000	-0.182	0.020
1077	YBR205W	KTR3	0	0.000	-0.182	0.020
1078	YKR061W	KTR2	0	0.000	-0.182	0.020
1079	YOR099W	KTR1	0	0.000	-0.182	0.020
1080	YPL061W	ALD6	0	0.000	-0.182	0.208
1081	YHR137W	ARO9	0	0.000	-0.182	0.036
1082	YLL052C	AQY2	0	0.000	-0.182	0.043
1083	YPR192W	AQY1	0	0.000	-0.182	0.043
1084	YGL202W	ARO8	0	0.000	-0.182	0.037
1085	YJL216C	IMA5	0	0.000	-0.182	0.036
1086	YIL172C	IMA3	0	0.000	-0.182	0.036
1087	YJL221C	IMA4	0	0.000	-0.182	0.036
1088	YGR287C	IMA1	0	0.000	-0.182	0.036
1089	YDL168W	SFA1	0	0.000	-0.182	0.091
1090	YMR170C	ALD2	0	0.000	-0.182	0.078
1091	YLR044C	PDC1	0	0.000	-0.182	0.015
1092	YGR087C	PDC6	0	0.000	-0.182	0.015
1093	YLR134W	PDC5	0	0.000	-0.182	0.015
1094	YFL018C	LPD1	0	0.000	-0.176	0.031
1095	YOR374W	ALD4	0	0.000	-0.182	0.064
1096	YGL256W	ADH4	0	0.000	-0.182	0.064
1097	YMR083W	ADH3	0	0.000	-0.182	0.064
1098	YBR145W	ADH5	0	0.000	-0.182	0.045
1099	YOL086C	ADH1	0	0.000	-0.182	0.045
1100	YDR242W	AMD2	0	0.000	-0.182	0.130
1101	YHR094C	HXT1	0	0.000	-0.182	0.000
1102	YDR345C	HXT3	0	0.000	-0.182	0.000
1103	YHR096C	HXT5	0	0.000	-0.182	0.000
1104	YJL214W	HXT8	0	0.000	-0.182	0.000
1105	YDR343C	HXT6	0	0.000	-0.182	0.000
1106	YMR011W	HXT2	0	0.000	-0.182	0.000
1107	YMR110C	HFD1	0	0.000	-0.182	0.008

F External tools and dataset manifest

Honest counts: 17 external tools and 10 accession-level datasets (`results/external_tools_manifest.csv`, `results/dataset_manifest.csv`). Dataset counting follows the program rule: one study's condition matrix is one dataset (Biolog = 1 dataset with 383 conditions; each chemostat compendium = 1). The manifests list every tool and dataset with version/accession, condition counts, and where each is used.